

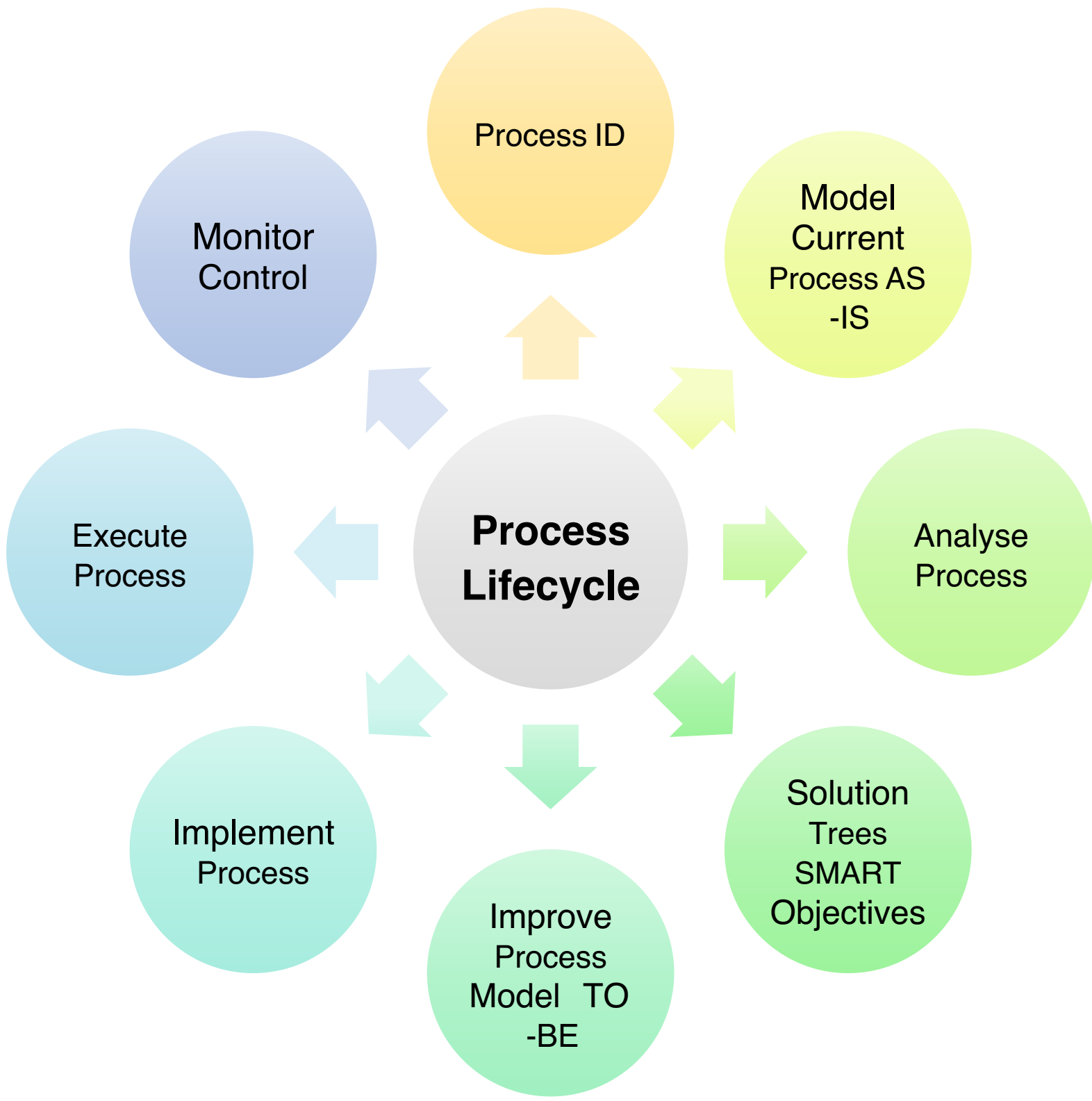
INFOSYS 704

IT Consultancy

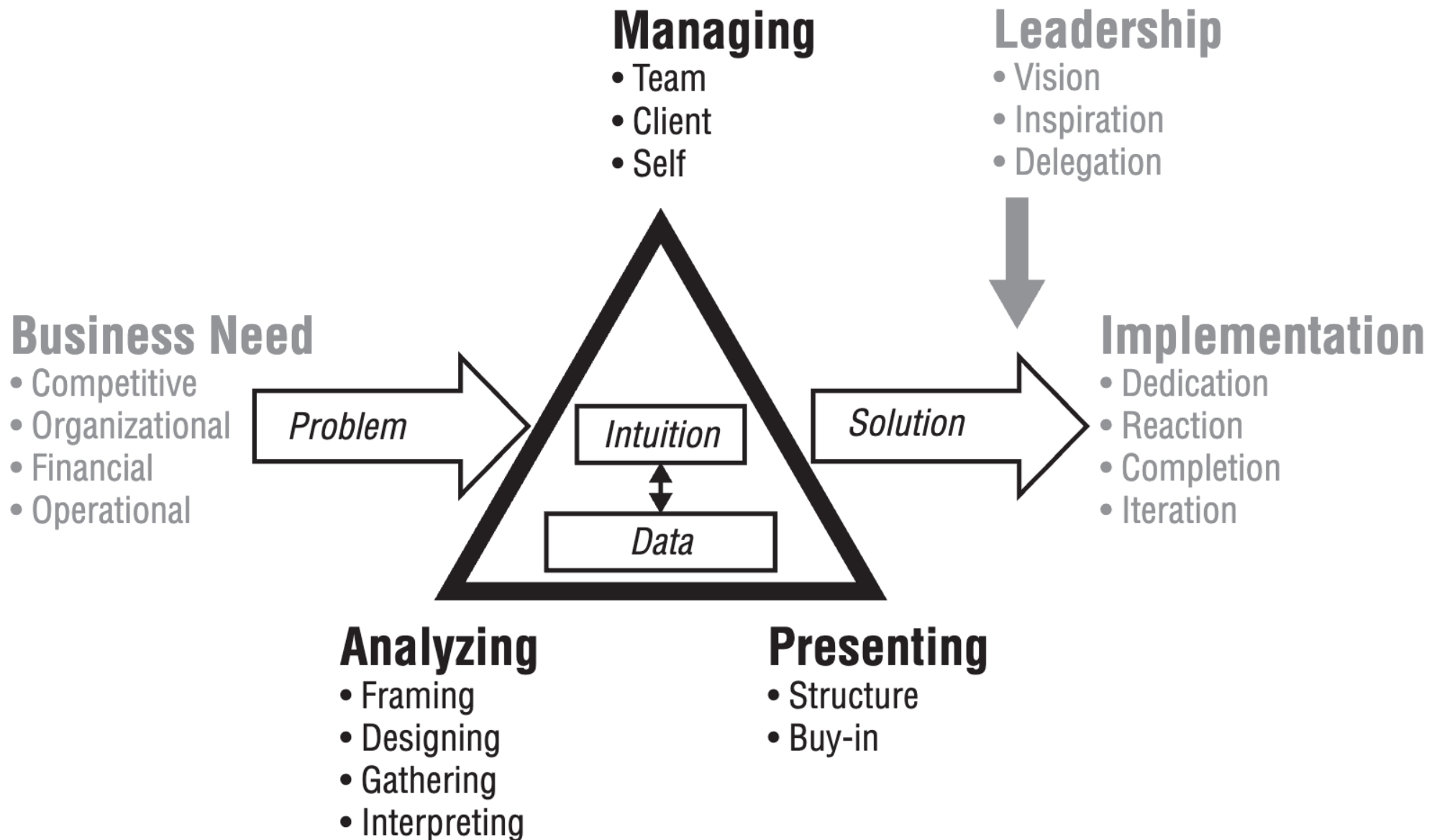
Lecture 6

Process Lifecycle





McKinsey Problem Solving Model



Process Lifecycle

The Process Lifecycle

- Understanding the Business
- Process identification
- Process modelling (as-is)
- Process analysis
- Process improvement (to-be)
- Process implementation
- Process execution
- Process monitoring/controlling



Rosemann (2001)

Process Identification

APQC Framework (2005)

OPERATING PROCESSES

1.0
Develop
Vision
and
Strategy

2.0
Design and
Develop
Products and
Services

3.0
Market
and Sell
Products and
Services

4.0
Deliver
Products
and
Services

5.0
Manage
Customer
Service

APQC Framework (2005)

MANAGEMENT AND SUPPORT SERVICES

6.0 Develop and Manage Human Capital

7.0 Manage Information Technology

8.0 Manage Financial Resources

9.0 Acquire, Construct, and Manage Property

10.0 Manage Environmental Health and Safety

11.0 Manage External Relationships

12.0 Manage Knowledge, Improvement, and Change

Gartner's Six Process Areas

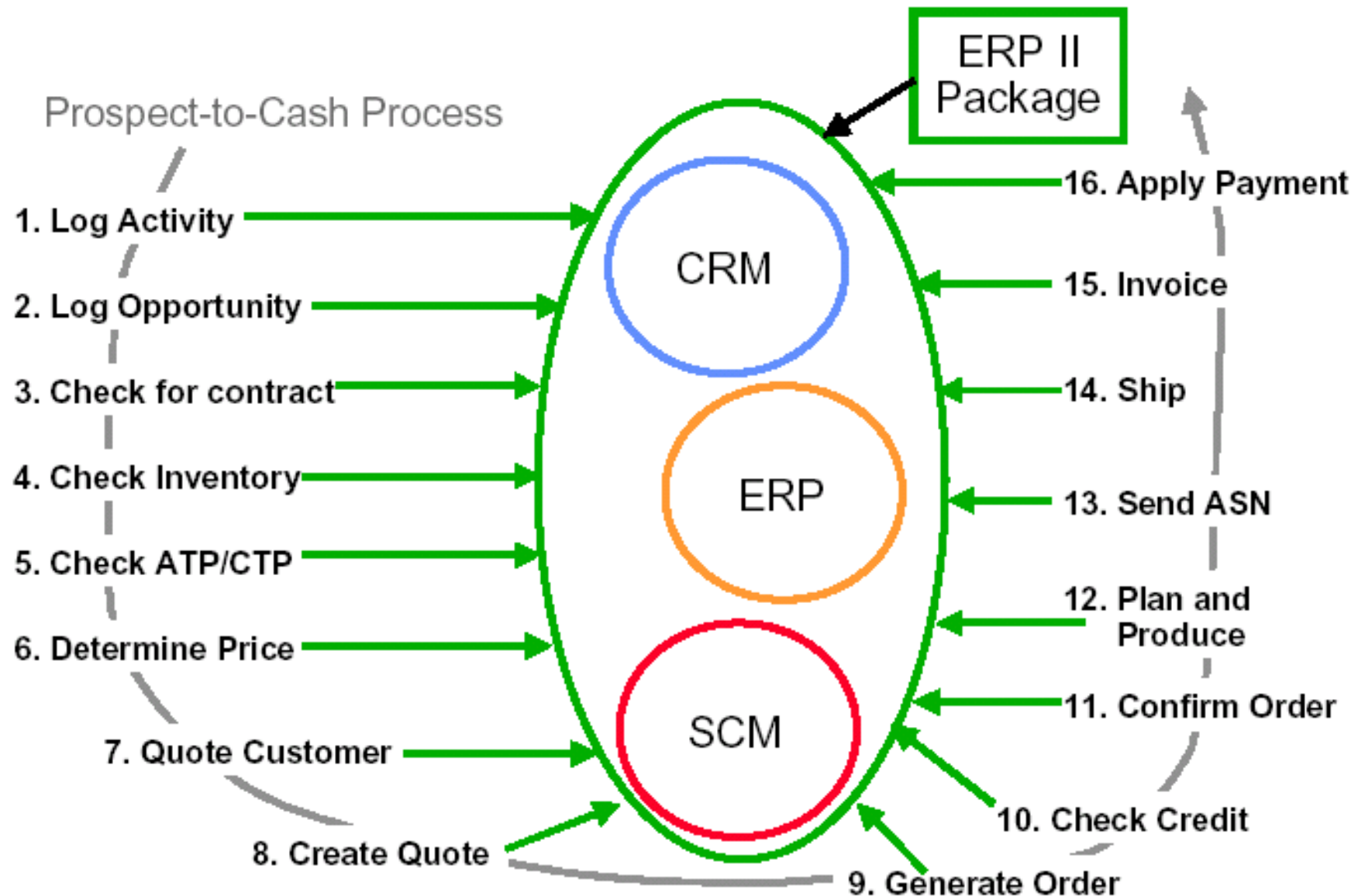
- **Prospect to Cash and Care:** This area encompasses the sales-order-to-cash process, bracketed by sales opportunity (i.e., prospect) management and post-sales service (i.e., care).
- **Requisition to Payment:** This area encompasses the procurement process and includes e-procurement capabilities.
- **Planning and Execution:** This area includes processes for planning related to resources (e.g., materials, cash, labor, transportation, personnel and maintenance), executing plans and controlling the resources used in plan execution.

Gartner's Six Process Areas

- **Plan to Performance:** This area includes financial processes that go beyond payables and receivables, to include consolidation, budgeting, treasury and financial reporting.
- **Design to Retirement:** This area encompasses processes associated with the product/service lifecycle, from conception to retirement.
- **Human Capital Management (HCM):** HCM processes support workforce-related issues for both human resources (HR) and non-HR business processes.

Prospect-to-Cash Process

(Gartner, 2001)



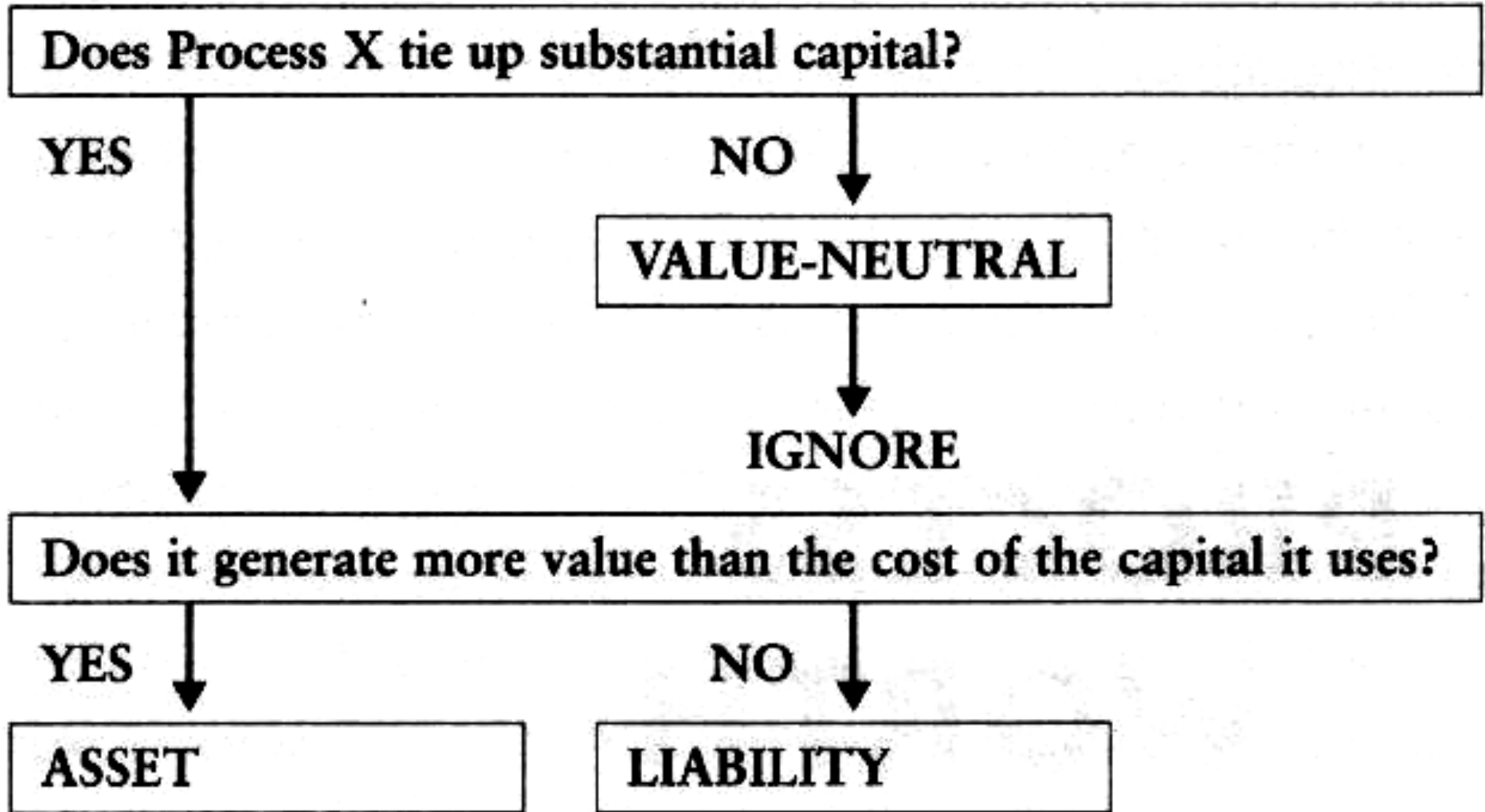
ASN
ATP/CTP

Advance ship notice
Available to promise/
capable to promise

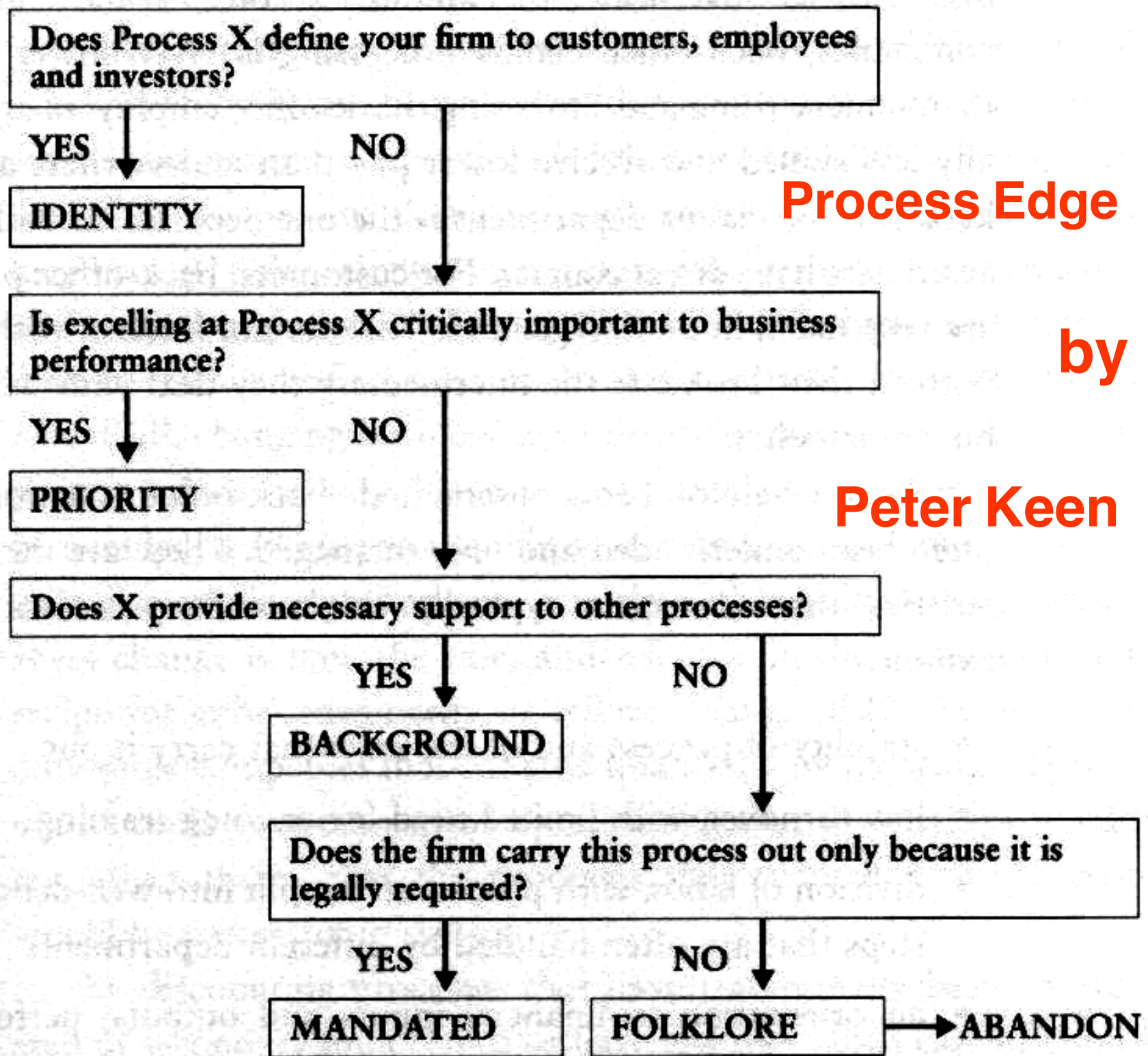
CRM
ERP
SCM

Customer relationship management
Enterprise resource planning
Supply chain management

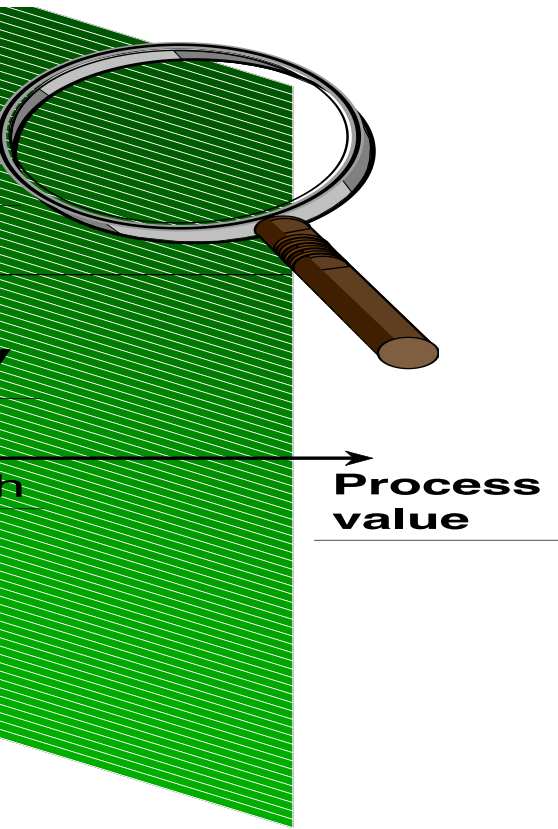
Determining Process Worth



Evaluating the Process Portfolio



Focus of the process selection



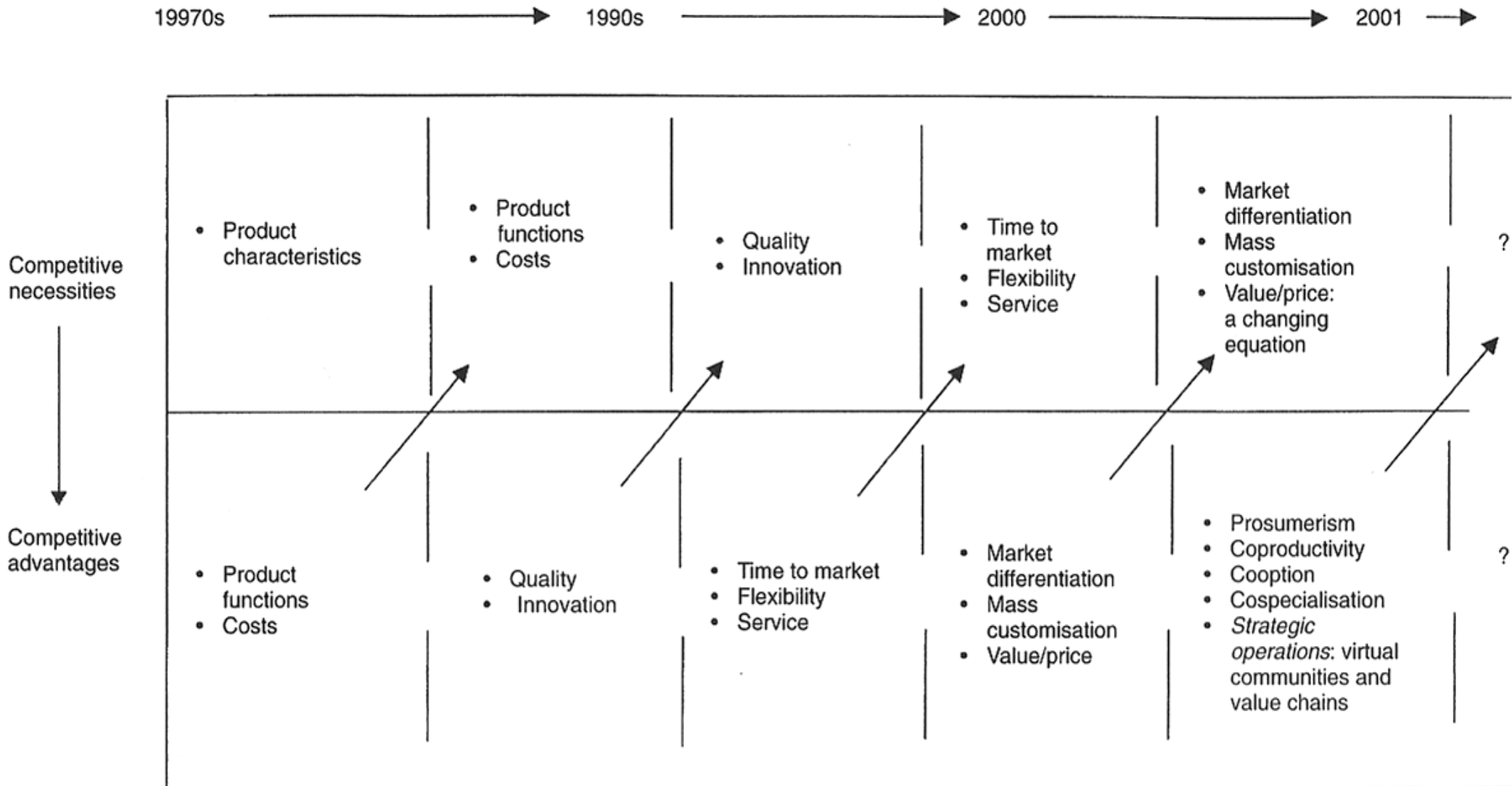
Hammer & Champy's (1994) Criteria

- dysfunction
 - "Which processes are in the deepest trouble?"
- importance
 - "Which processes have the greatest impact on the company's customer?" and
- feasibility
 - "Which process is the most susceptible to successful redesign?"

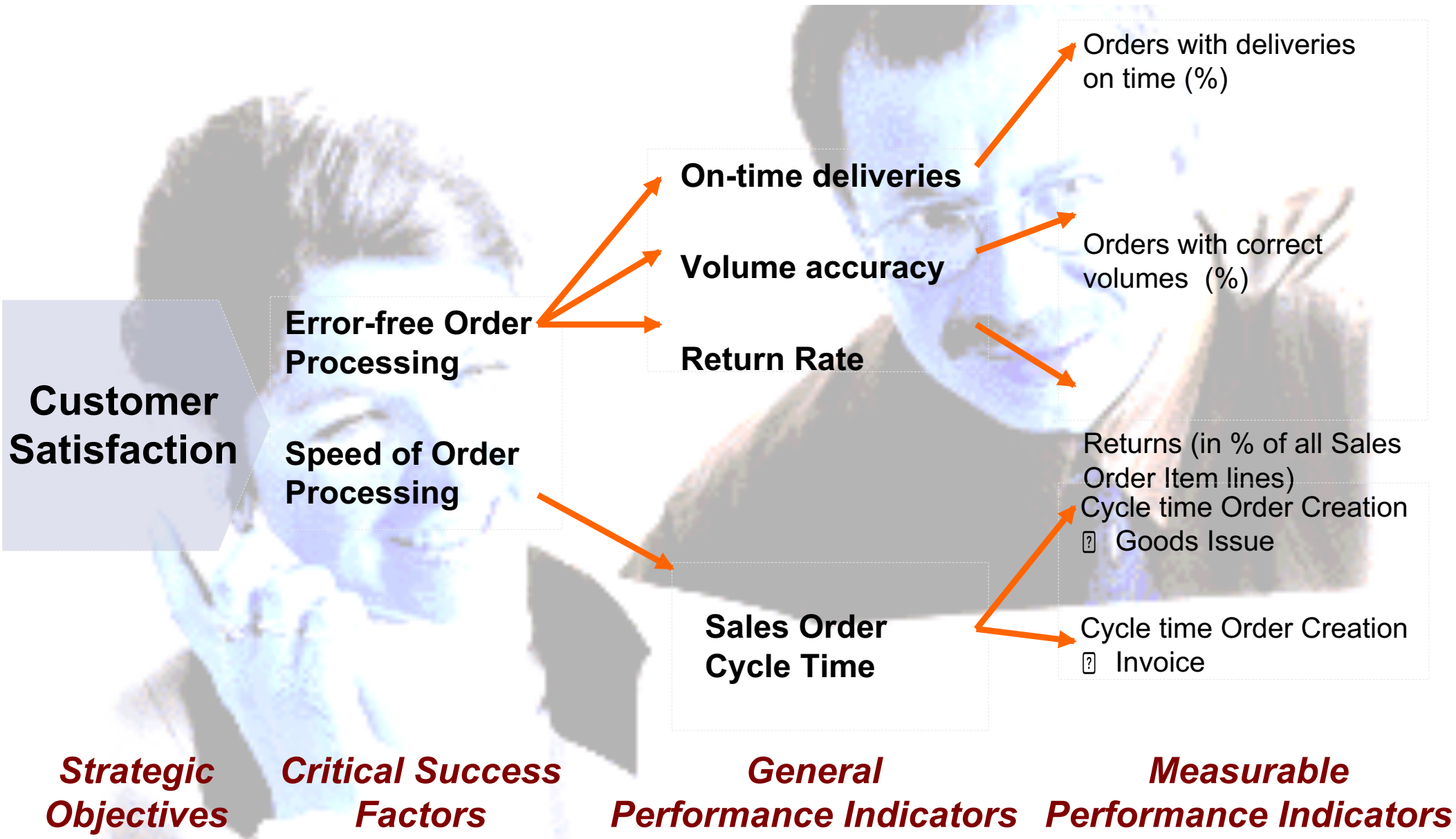
Davenport's (1993) Criteria

- enumerate major processes
- determine process boundaries
- assess strategic relevance of each process
- render high-level judgements of the 'health' of each process
- qualify the culture and politics of each process.

Stages of Competitiveness (Walters, 2002)

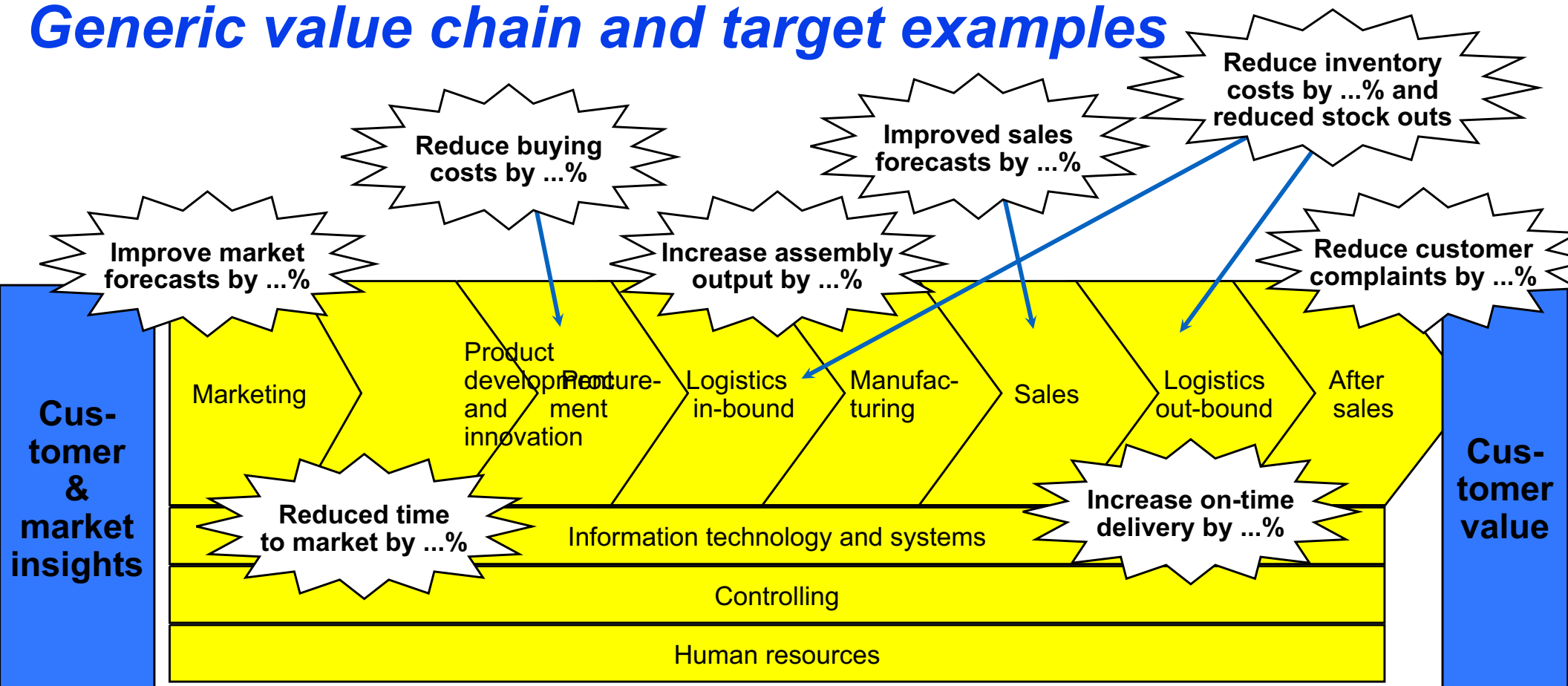


Strategic Goals => KPI's



The targets are presented along the value chain to identify direct interrelations and target contributors

Generic value chain and target examples



Analysing the processes which contribute to the overall targets along the value chain is key to integrate all corporate departments and process owners into the ERP project.

KPI's => Processes (SAP)

Benchmarking to Identify Processes

Historical

Compare KPIs results with historical results, i.e. year 1997 vs. 1998

Internal

Compare KPIs results within your company, i.e. profit centers, departments

Benchmarking

Theoretical

Analyze the circumstances, set yourself targets, compare your targets with your KPIs results

External

Compare KPIs results with other companies of the same business

SAP + PMG Alliance

Goal /
Vision

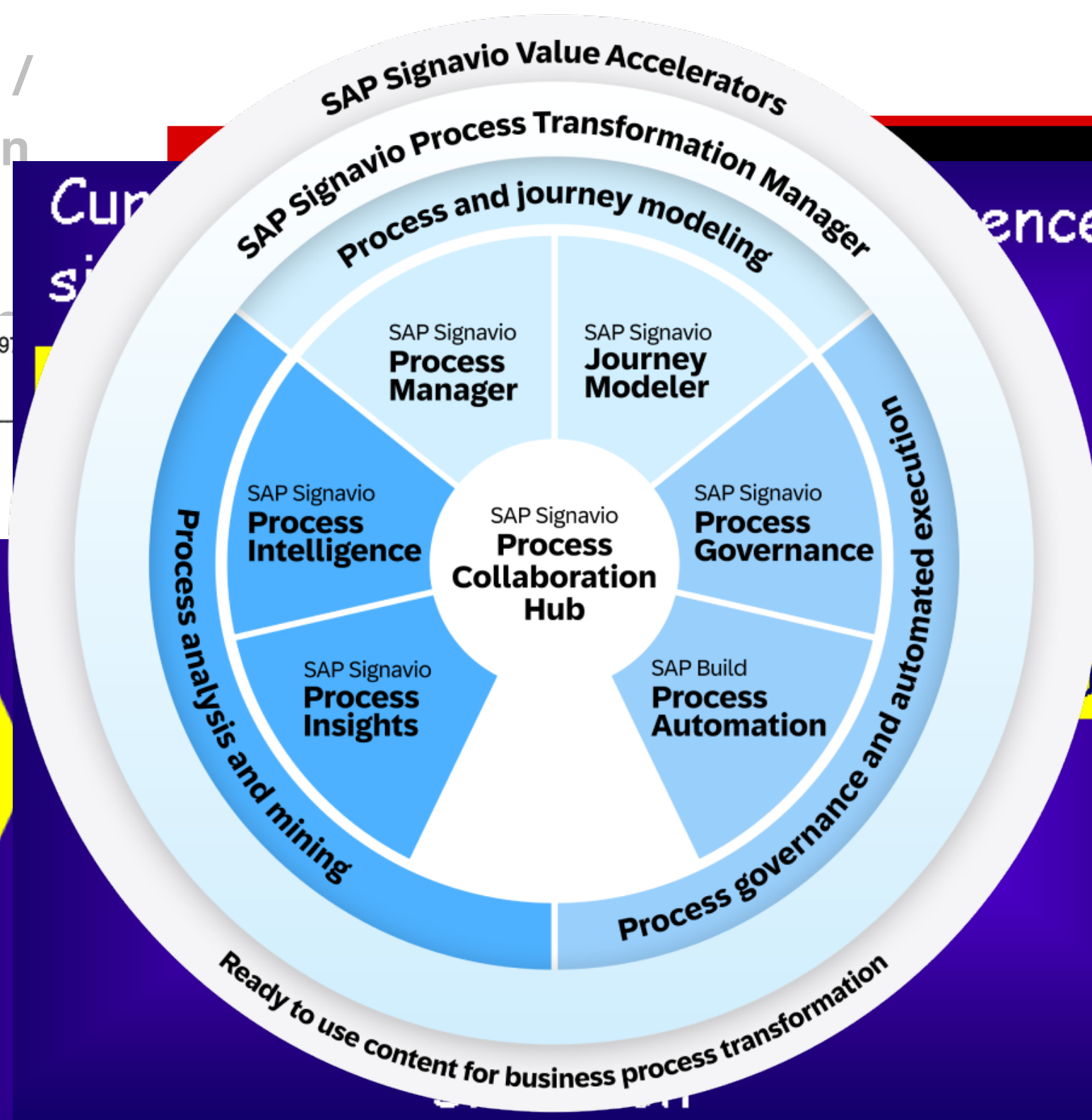
Current
State

Performance

Best
Business
Practices

Customer
Satisfaction

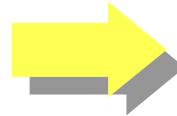
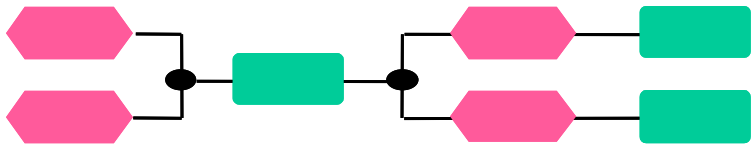
Strategic
Objectives



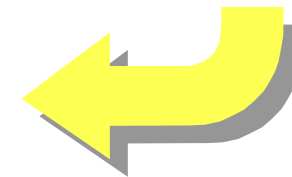
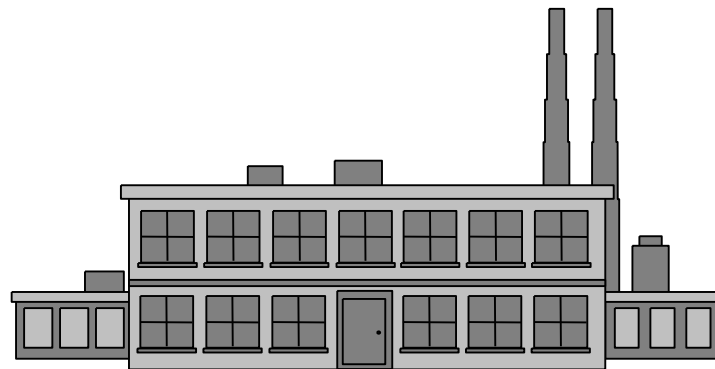
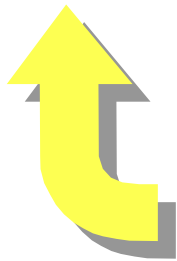
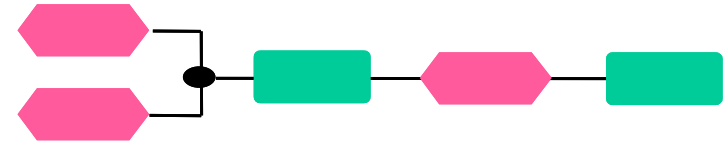
AS-IS Process Modelling

Objectives of Process Modelling

**Descriptive modeling
of the real world (as-is)**



**Prescriptive modeling
of the real world (to-be)**

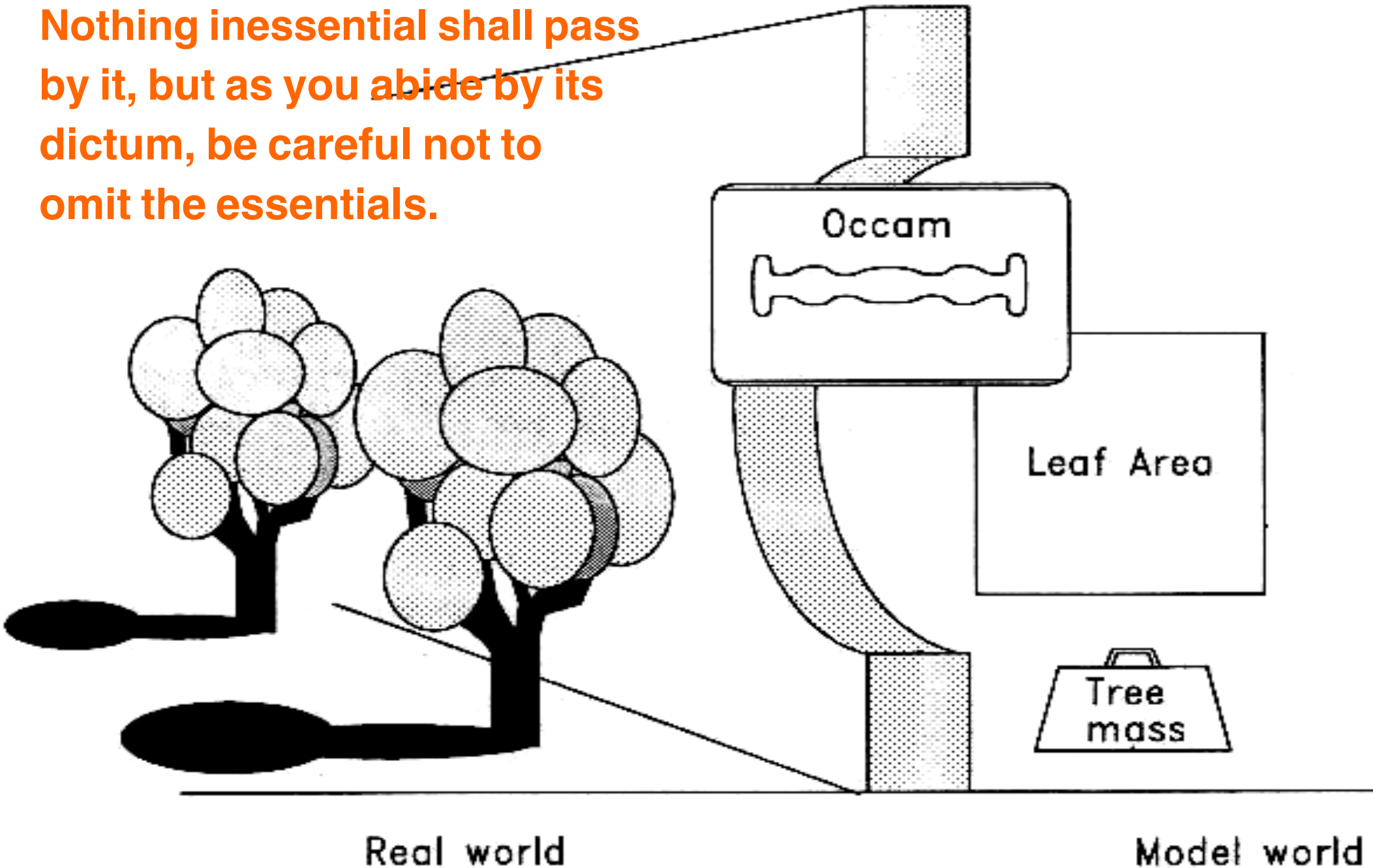


Document Business processes

- “Staple yourself to an order” (Shapiro et al, 1992)... or a customer, student, employee, manager, enquiry, raw material ...
- An example scenario for an order could be: order planning, order generation, cost estimation and pricing, order receipt and entry, order selection and prioritisation, scheduling, fulfilment, billing, returns and claims, post-sales service.
- Study the whole system by stapling yourself to the business object – travel horizontally through the organisation... Work in different roles within the organisation

William of Ockham's Razor – 1200 AD

Nothing inessential shall pass by it, but as you abide by its dictum, be careful not to omit the essentials.



Pros and Cons of Process Modelling

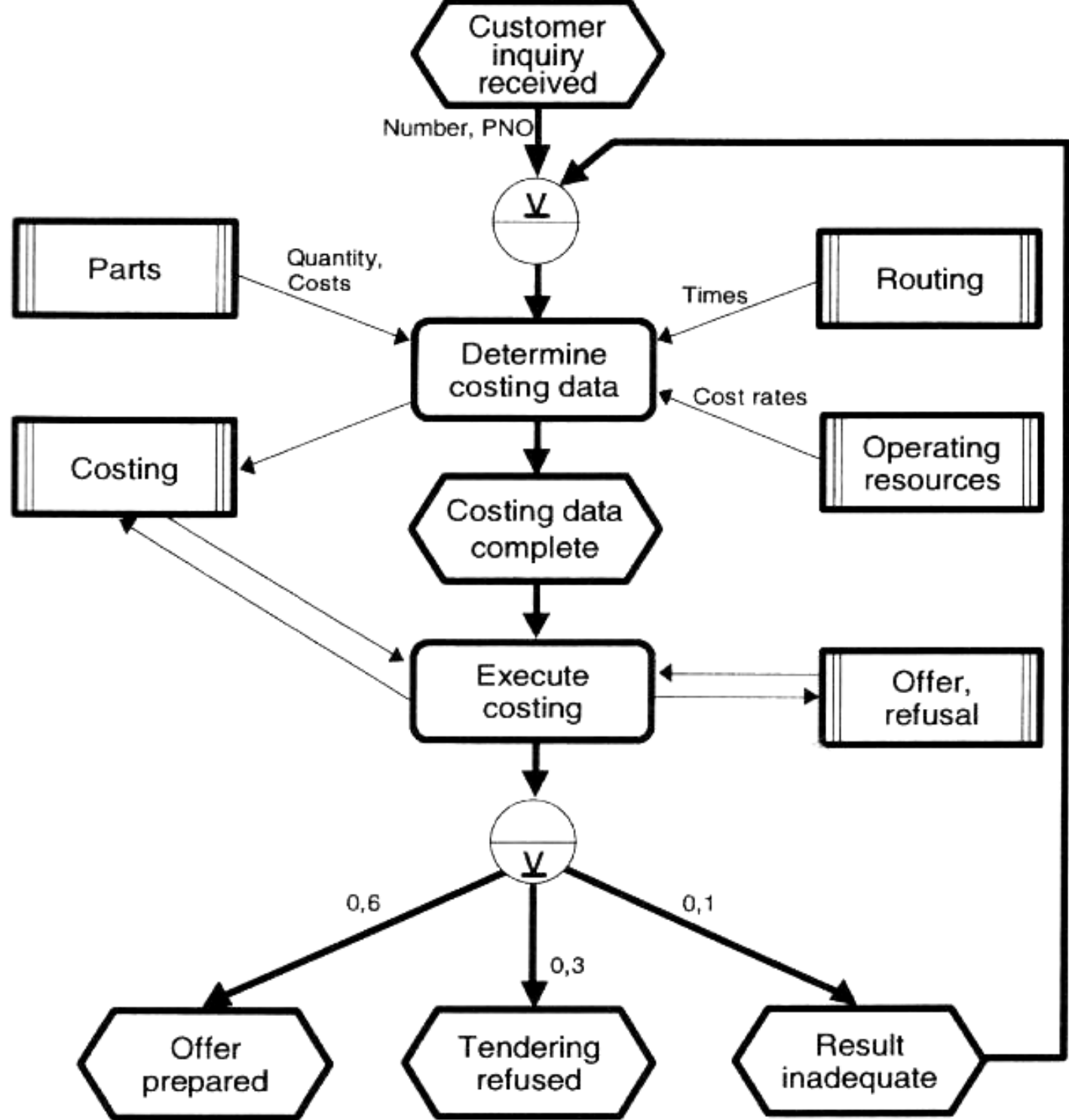
Advantages

- ? the same problem of understanding
- ? a defined project scope
- ? the same terminology
- ? it supports acceptance for the project (unfreezing)
- ? the base for a migration strategy towards the redesigned processes
- ? the completeness of to-be processes can be evaluated
- ? the results of as-is analysis can be used as to-be, if there are no or only minor changes
- ? it shows weaknesses and restrictions

Disadvantages

- ? the results are obsolete as soon as to-be processes are designed and/or implemented
- ? the danger of narrow focused process design (thinking in constraints)
- ? time- and cost-consuming

Order Processing with Data Flows



Legend:

→ Control flow

→ Data flow

Reasons for Modelling	Aspects to Consider
Business Planning	Concentrate on business objectives, customer needs and metrics. Use <i>Value Chain Diagrams</i> , <i>Balanced Scorecard Models</i> . Look at high-level business functional breakdown.
Business Restructuring	Concentrate on the organisations that are carrying out tasks and the hand-offs between them. Look at the value added by each Function. Use <i>eEPCs as row display</i> to provide organisational swim-lane view.
Baseline the Business	Usually impractical in all but the simplest and static businesses. It takes too much time and the world moves on in the meantime. Concentrate instead on key processes, processes that have to change or where you already know there are problems. "Don't model the Universe".
Operational Process Design	Concentrate on getting the flow of the process correct. Use <i>eEPCs</i> and <i>FADs</i> . Identify the key decisions being made. Look for failure paths as well as the normal process flow. Identify inputs and outputs for all tasks. Identify key documents and sources of information.
Systems Development	Requires very detailed logic flow to be modelled. Exception handling is very important. Detailed data models, data flow and systems interfaces must be modelled.

Modelling Viewpoint	Approach
Follow a data object	Possibly the easiest approach to take. Select a key data item (e.g. a customer order) and follow it through the process. See what actions are performed on it, who handles it and where it ends up. This is also useful for testing other model viewpoints.
Model the business	Modelling what the whole business does is one of the hardest approaches. It can normally only be done at high levels of abstraction and it is often difficult to identify the triggers and outcomes.
Model a business function	The most common approach is to model a particular business function (e.g. order-handling, fault-reporting, etc.). This will normally involve many different organisational units. Modelling organisational hand-offs will be essential. Can lead to very 'company oriented' models that forget about the customer.

Modelling Viewpoint	Approach
Model a business process	The most useful approach (but not often done) is to model the end-to-end processes that a business function performs. Particularly valuable when done from a customer perspective. Better than the business function approach as it helps ensure the whole process hangs together to deliver a good customer experience. Helps identify failure modes.
Model an organisation	Another common approach is to model what an organisation does. This may not necessarily be the most useful approach. Organisations change over time and the range of functions an organisation performs may have evolved historically.
Model an organisational unit	This model just focuses on what a single unit does. The model shows the interfaces with other units, but doesn't worry about how they accomplish their tasks. Provides a very focused model, but can over-simplify what is going on. It may also encourage an out-of-sight, out-of-mind approach, which doesn't spot gaps and failure points in the end-to-end process.

Modelling Viewpoint	Things to Consider
Business Unit	Business units (e.g. Sales, Manufacture, etc.) provide a pragmatic modelling abstraction. They are well understood and have major significance to the business. They do not change frequently, but when they do they have a significant impact on business processes which you have to reflect in your model.
Line Management Team	To be avoided at all costs. Have very little business significance, change frequently and have little impact on processes when they do.
Operational Centre	A very useful modelling abstraction. Normally have a very significant impact on process, change infrequently and fit within Business Units. Consider whether it is sufficient just to nominate the operational centre that does a task (e.g. Sales Office) or whether it is necessary to be more explicit about the roles within the centre (e.g. Sales Office Customer Service Advisor).

Modelling Viewpoint Management Layer

Roles

Things to Consider

Many business sectors have functional layers. For instance in Telecommunications we talk of Service Management, Network Management, etc. They can provide a useful level of abstraction, but can be problematic because often there is no clear definition of what they mean. Useful when modelling 'to-be' scenarios, but Business Units are better for 'as-is' models.

The lowest practical level of organisational unit. If used in process models you should make sure they are unique. The role "Customer Service Advisor" or "Planner" when attached to a Function doesn't convey very much as many operational centres will have such roles. You can model their parentage in an organisation chart, but better to make the names unique and meaningful (e.g. Sales Office – Customer Service Advisor).

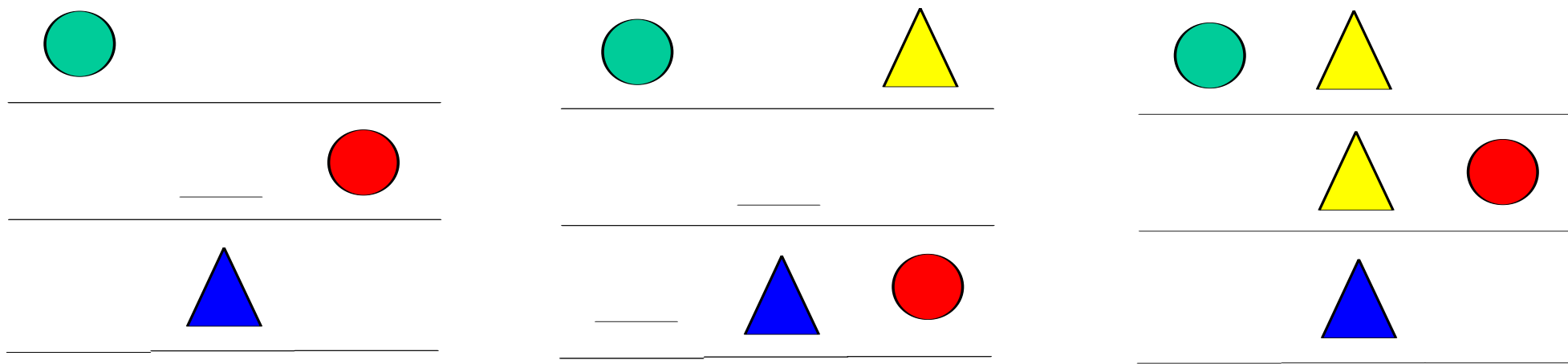
Modelling Viewpoint	Things to Consider	
'as-is'	The process as it is now. But be careful to define what 'now' means. Does it mean as documented now or as operated now? If you know the documented process is wrong and a 'work-around' is being used, you have to decide which you are going to model. If changes are being made while you are capturing the process, are you going to include them or freeze the model at a certain point in time?	
'to-be'	How far in the future are you considering? Are you starting from scratch with a new business model or re-engineering what you already have? How radical can you be? What constraints have you got?	
Time-scale	Complex processes can sometimes complete within seconds, while simple processes can often last for weeks. What time-frame is important to your model?	
Delays	Are potential delays in the process important to you? Populating Function attributes with process times may not make delays explicit to people viewing the model. Consider explicitly modelling delays as additional Functions if you want to draw attention to them.	
Simulation	Simulation can be used to analyse process delays and optimise performance. A very powerful tool, but requires good quality models and good quality data. Models must conform to certain rules. Consider seeking advice from specialist simulation experts.	Davis,2001

Process Analysis

Process Analysis

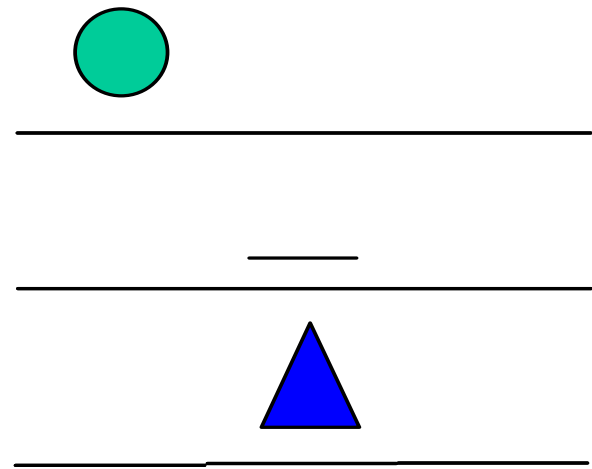
- What is the aim of the process?
- Why is the process executed as it is?
- Which organizational units and application systems are involved - and where are the interfaces between them?
- What are the current problems?
- Which changes independent from the Process Engineering project will take place soon (internal/environment)?
- What technology is (will be) available?
- What are the current benchmarks?

Reference Models – Generic Solutions



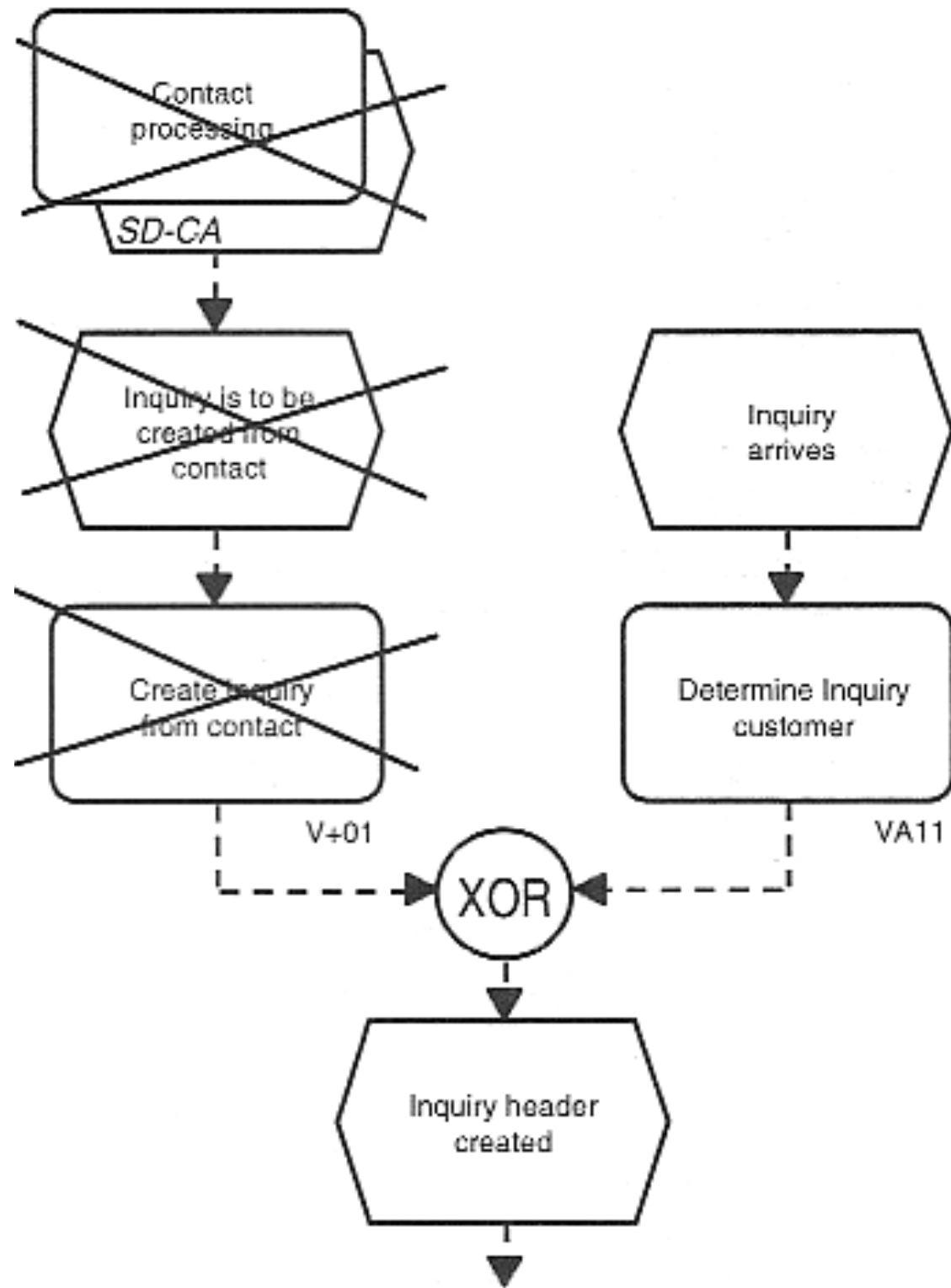
Three individual models

SAP has 800 business process models and 120 for collaborative business processes for e-commerce



Possible Reference model

Using the Reference Model as a starting point in process analysis



Solution

Trees

SMART

Objectives

Solution Trees

- A solution tree is a type of logic tree that focuses not just on breaking down a problem (as in an issue tree) but on mapping out potential solutions in a structured MECE way.
- A visual framework used to decompose a key problem statement into distinct solution pathways, ensuring all options are considered without overlap, & each branch is logically connected to the root problem.
- Forces consultants to think solution-first once the problem is well-defined. Ensures no critical solution path is missed. Facilitates prioritisation in workshops with clients. Helps in storylining final recommendations.

Solution Trees Characteristics

- **Root node** = Central problem statement (e.g., "How can Company X grow revenue by 15% in 12 months?")
- **First branches** = Main solution categories (e.g., "Increase sales volume", "Increase price", "Expand product lines")
- **Sub-branches** = Specific initiatives or actions under each category
- **MECE principle** = No overlaps between branches, and combined they cover all reasonable possibilities
- **Prioritization** = Often used to evaluate solutions by feasibility, cost, and impact

Solution Tree - Root Problem: “How can a New Zealand bank grow its digital customer base among Māori and Pasifika communities by 25% in 18 months?”

- **Enhance Digital Product Offering**
 - Develop culturally tailored mobile banking features
 - Introduce low-cost remittance service
 - Expand language options in app
- **Improve Accessibility**
 - Partner with community hubs for digital literacy programs
 - Offer zero-data usage on banking app
 - Create offline transaction capability
- **Targeted Marketing**
 - Use local influencers and storytellers
 - Sponsor community events
 - Localized advertising in Māori & Pasifika languages
- **Incentivize Usage**
 - Cashback for digital bill payments
 - Bonus interest for online savings accounts
 - Referral rewards

**Solution Tree - Root Problem: “How
can we grow retail store revenue by
20% in 12 months?”**

- **Increase Transaction Volume**
 - Expand store hours
 - Improve store layout for faster service
 - Increase marketing spend
- **Increase Average Transaction Value**
 - Upsell complementary products
 - Loyalty program upgrades
 - Bundling discounts
- **Expand Customer Base**
 - Open new locations
 - Online store launch
 - Partner with delivery platforms

**Solution Tree - Root Problem: “How
can the bank improve customer
experience through digital
transformation?”**

- **Enhance Digital Platforms**
 - Mobile-first redesign
 - AI-driven chatbots
 - Omnichannel integration
- **Streamline Operations**
 - Automate loan approval
 - Digitize paper forms
 - RPA for customer service
- **Data-Driven Personalization**
 - Targeted offers based on spending
 - Predictive fraud detection
 - Customer lifetime value modeling

“How can we reduce manufacturing costs by 15% without hurting quality?”

- **Process Efficiency**
 - Lean manufacturing
 - Reduce machine downtime
 - Better shift planning
- **Supply Chain Optimization**
 - Negotiate supplier contracts
 - Bulk purchasing
 - Nearshoring suppliers
- **Waste Reduction**
 - Recycle production scrap
 - Energy efficiency upgrades
 - Improve quality control to reduce defects

“How can we design an app to improve Indigenous health outcomes?”

- **Cultural Relevance**

- Language localization (Māori, Pasifika languages)
- Indigenous art and design
- Elders as trusted messengers

- **Functionality**

- Symptom tracking
- Appointment booking
- Medicine reminders

- **Community Engagement**

- Peer support forums
- Local event calendars
- Health challenges with rewards

Solution Tree - Root Problem:
“How can we reduce customer churn by 10% in 12 months?”

- **Improve Customer Service**
 - Faster call center response times
 - AI-powered self-service portal
 - Proactive issue resolution
- **Enhance Product Offering**
 - Better mobile data packages
 - Value-added streaming services
 - Flexible payment plans
- **Strengthen Customer Engagement**
 - Loyalty rewards program
 - Personalized offers based on usage
 - Targeted win-back campaigns

SMART Objectives

- S = Specific (not: make better products)
- M = Measurable (not: maximum profit)
- A = Acceptable (not: eliminate competitor X with all means)
- R = Realistic (not: double the turnover within 3 months)
- T = Time related (not: reduce costs for 30%. When ever that may happen)
- KPI's need to be Specific, Measurable, and Acceptable

Customer Satisfaction

Success Factor	Suggested PI
Improved customer relationships	% of communications not returned by company
	Negative responses from site meetings
	Number of contacts with customer during project and post-project success
	Number of visits made

Increase Customers

Success Factor	Suggested PI
Getting a larger slice of business from our important customers	% of top 10 customers business
	Top 10 clients measure % of dollar value business company has - number of projects

Employee Satisfaction

Success Factor	Suggested PI
Retention of senior staff	Staff turnover especially senior staff
Learning from projects	Number of post project reviews /debriefs
Increase cross skilling	NMAP
Overseas and “fun” factor	Number of days working overseas on jobs

Financial

Success Factor	Suggested PI
Improved cashflow	% of those paying cash up front
	% of new projects where payment up front received
	Costs of lost tenders Vs costs of won tenders
	Timing of billing on milestones e.g. if 5 days late result would be + 5 days etc
Measure success of tenders	% of successful tenders
	% of successful tenders a project group gets
	Number of customers who increase consulting by xx% per quarter

Learning and Growth

Success Factor	Suggested PI
Rising stars	No. of Professional leadership initiatives to targeted rising stars
Up skilling marketing and selling	Number of relationships producing real work
Mentoring is the key link for both young and old	% of staff in mentoring relationships
Up-skilling (continual) of junior staff "knowledge growth"	% of rising stars with mentors
	Hours in mentoring,
	Number of mentoring meetings by rising star
Training (Achieving CPD hours)	% of staff meeting CPD requirements
Regular performance reviews	% of reviews completed on time

Process Improvement

TO-BE Modelling

Process Improvement / Transformation

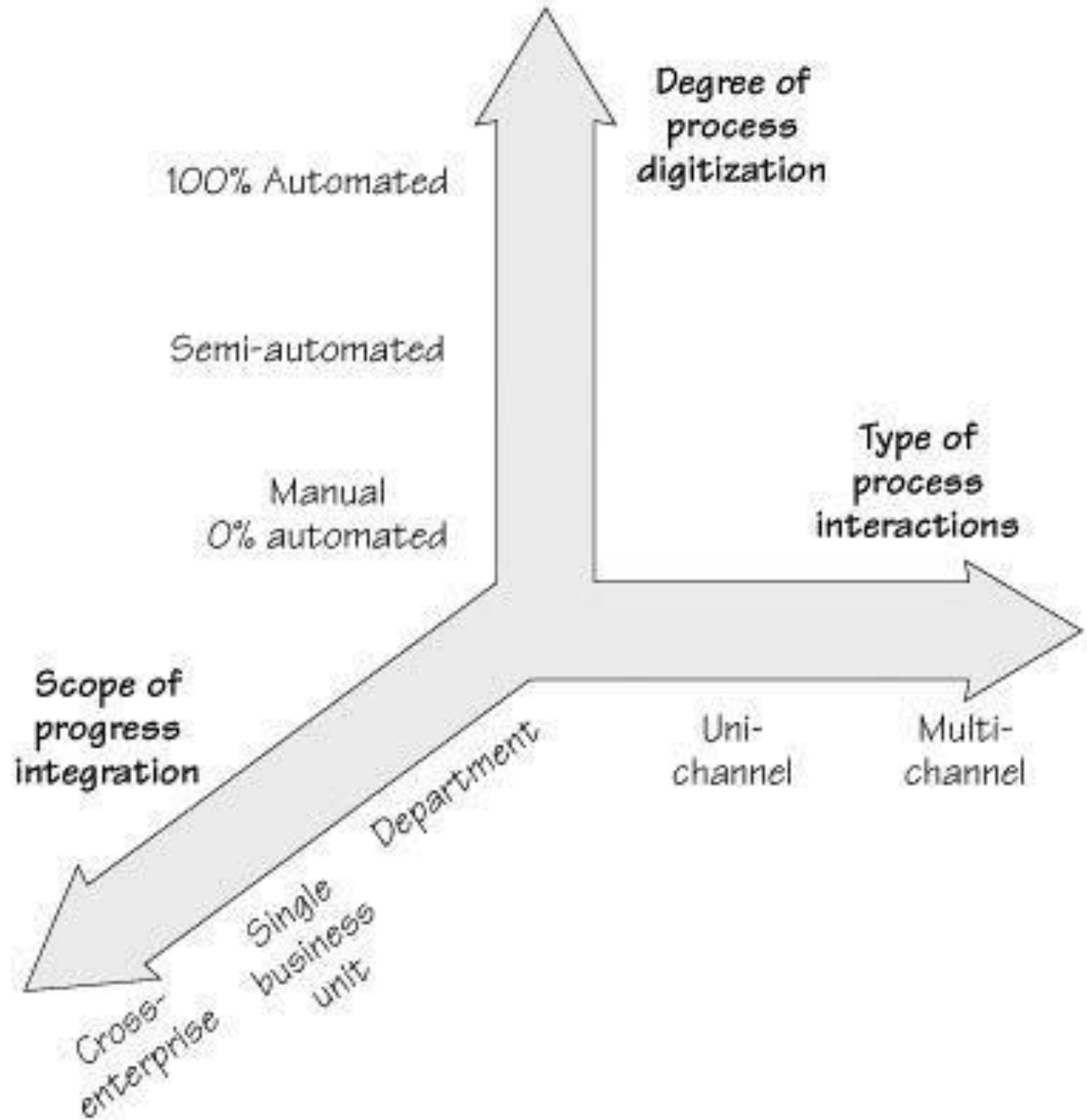


Figure 1.1: Process Transformation Dimensions

Objective	Example	Description
Process Improvement	GE Honeywell Caterpillar	- Reducing hand-off costs and rework - Reducing process variation using Six Sigma - Transaction cost reduction—using technology to shorten end-to-end processing time - Business process outsourcing—fixing specific errors - Improving customer satisfaction
Strategic Improvement	Wal-Mart Dell Intel Cisco	- Replacing manual processes with digital processes wherever possible - Enhancing logistic and supply chain efficiencies - Increase customer focus—Web-enabling strategic operations - Cycle time reduction—decreasing time to market
Business Transformation	IBM AT&T Vivendi	- Rewiring the industry—changing the rules of the game - Creating a new customer focus—from product-centric to customer-centric - Abandoning old ways of doing business - Major culture change required among all participants

Figure 1.3: Three Categories of Execution

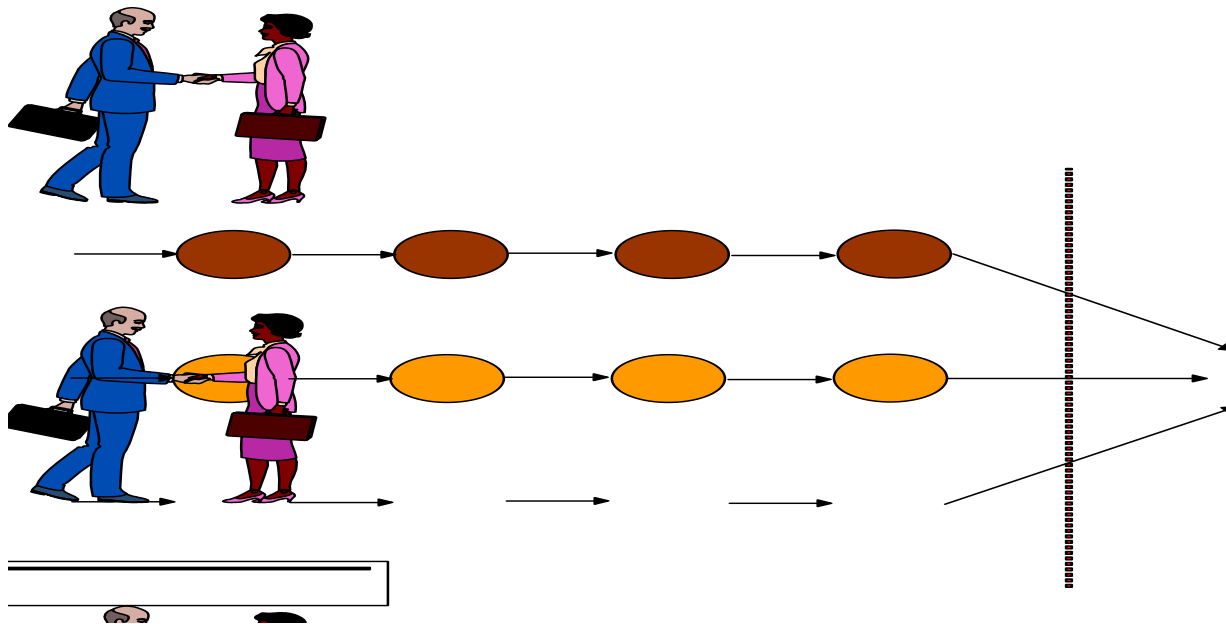
Process Improvement

- improvements related to the specific *outcome* of a process
- improvements related to the flow of *activities* of a process
- improvements related to *resources* involved in a process

Outcome-related Improvements

- Elimination of an Outcome
- Substitution of an Outcome
- Digitalization of an Outcome
- Harmonization of an Outcome – horizontal (standardisation of documents or products) and vertical (standardisation and integration of processes)
- Separation of Outcomes (reverse of harmonisation)
- Process-oriented Optimization an Outcome

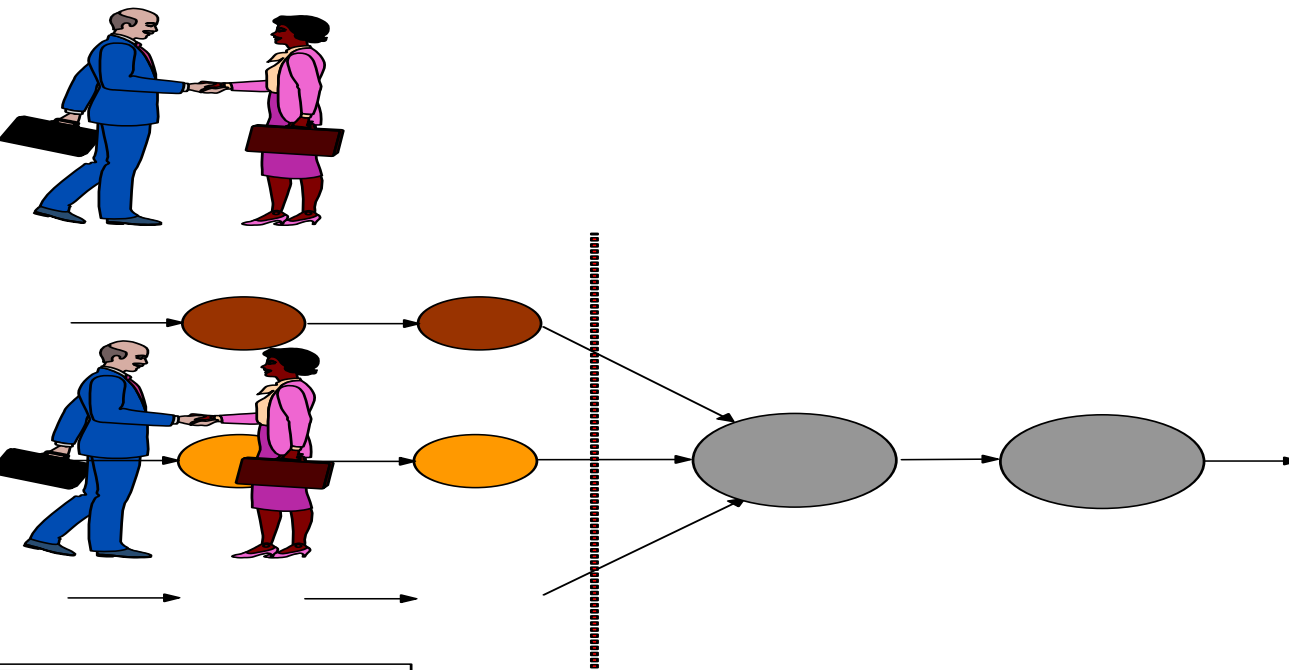
Standardization of processes (1/4)



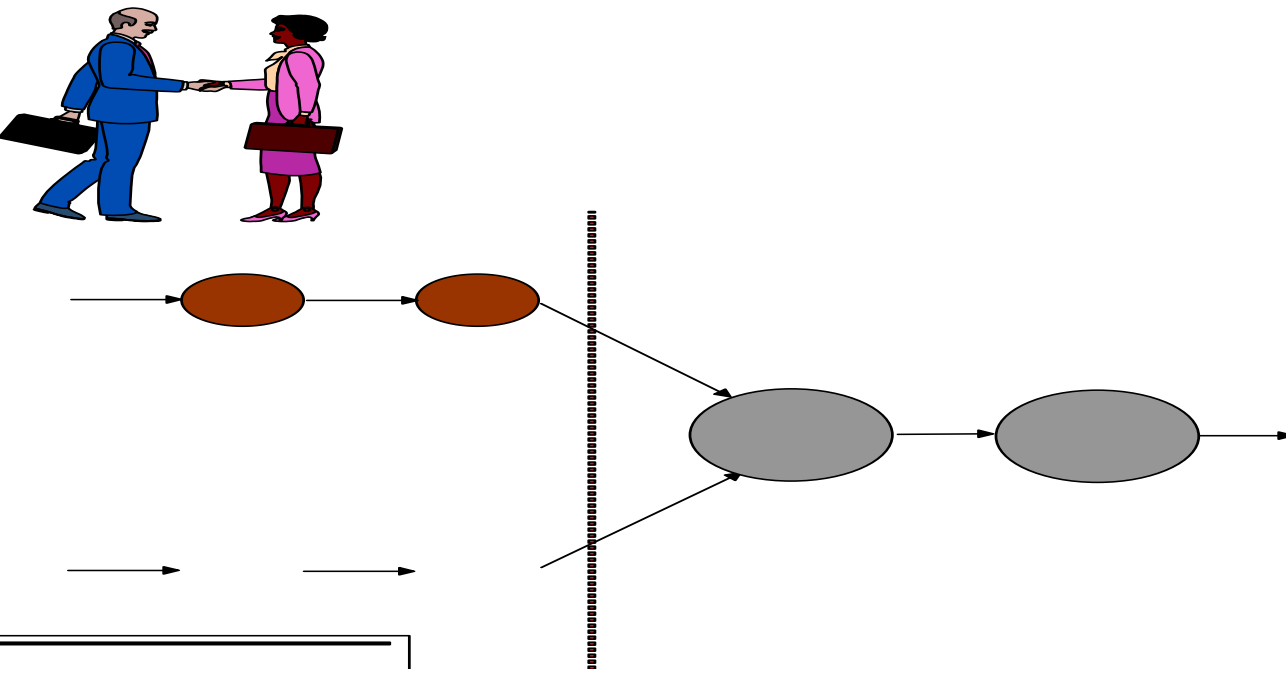
Line of Visibility

Rosemann (2001)

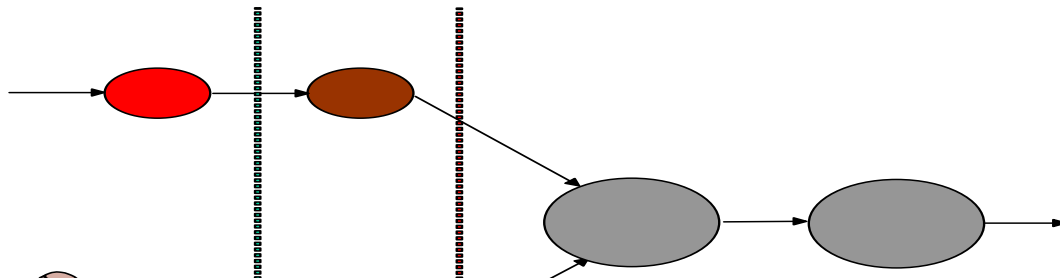
Standardization of processes (2/4)



Standardization of processes (3/4)

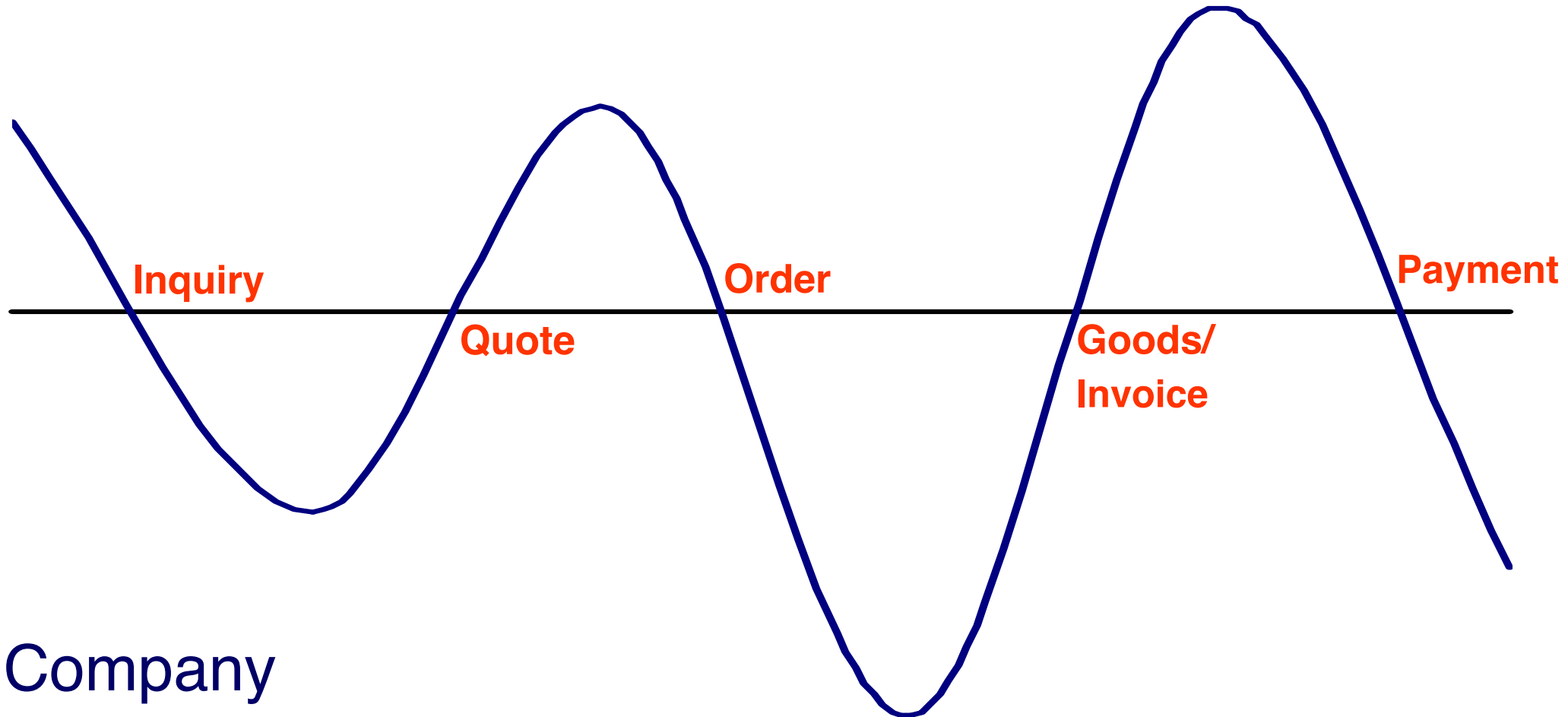


Standardization of processes (4/4)



Visibility of Customer Interaction Cycle

Customer

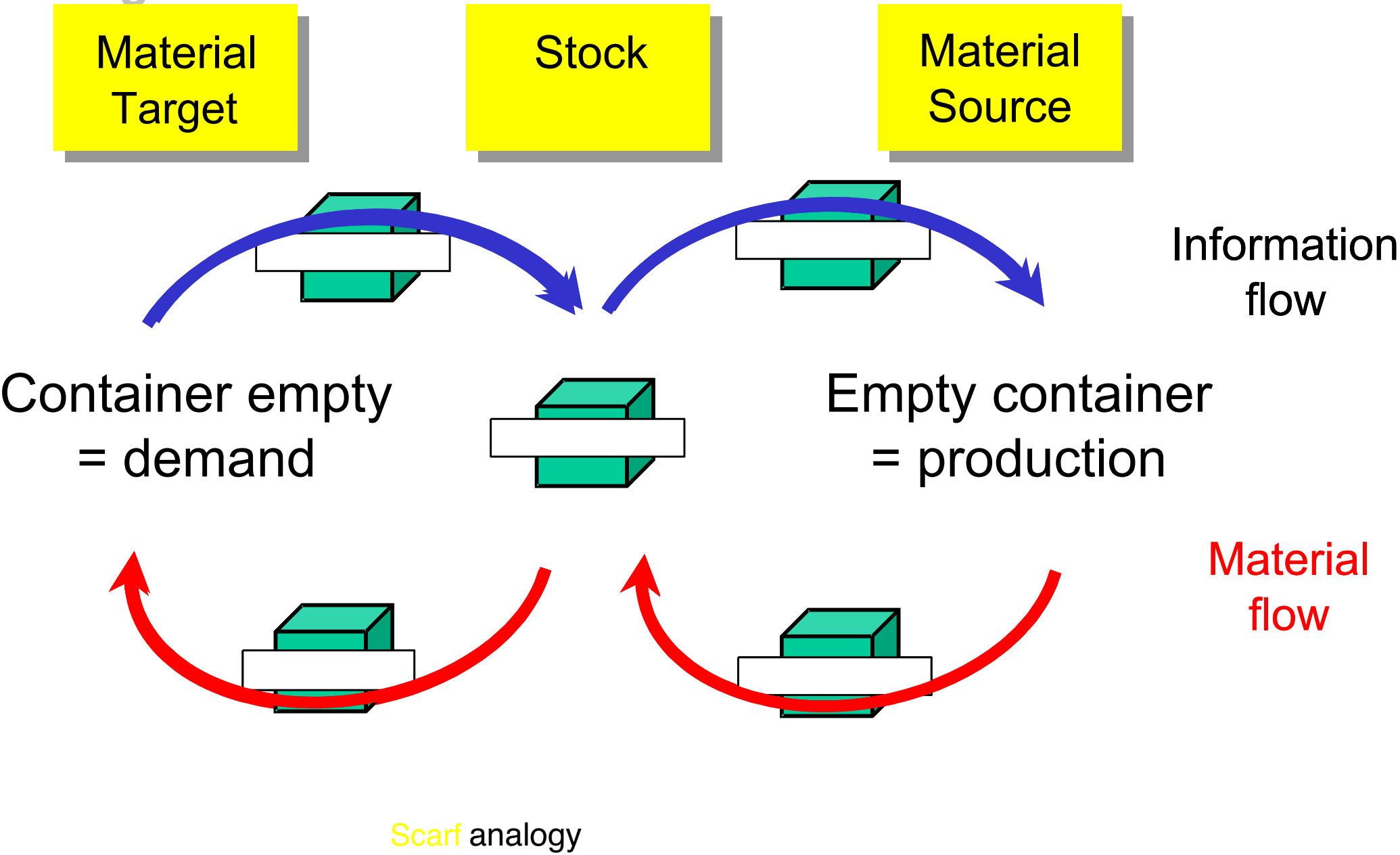


Activity-related Improvements

- Elimination of Activities
- Change the Flow of Activities: Pull Instead of Push
- Focus on Bottleneck Activities
- Automation of Activities
- Parallel Routing of Activities

eliminate
↓
simplify
↓
Automation.

Change the Flow of Activities: Pull Instead of Push



Focus on Bottleneck Activities

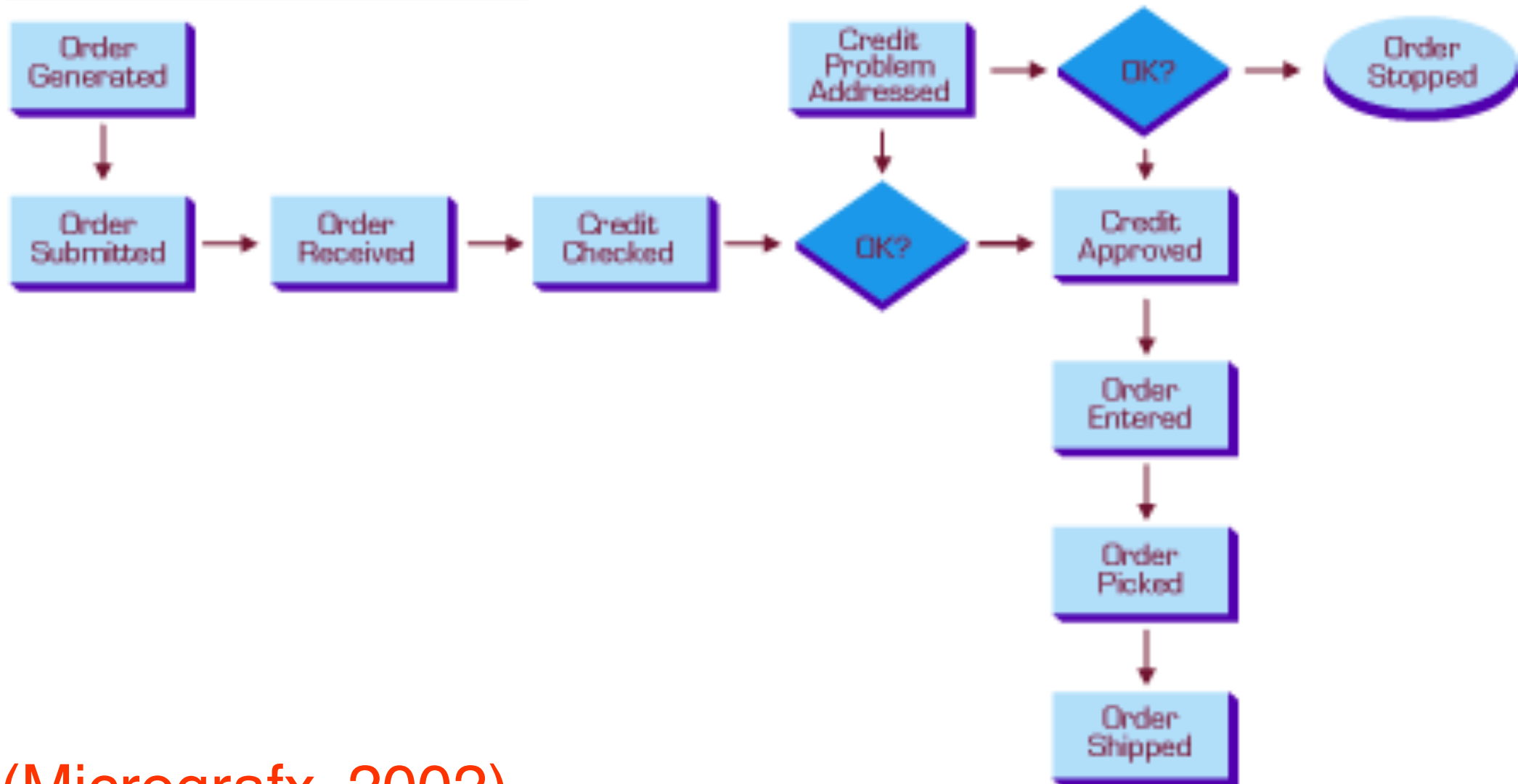
- Balance flow, not capacity.
- The level of utilisation of a non-bottleneck resource is not determined by its own potential, but by some other constraint in the system.
- Utilisation and Activation of a resource are not the same.
- An hour lost at a bottleneck is an hour lost for the entire system.
- An hour saved at a non-bottleneck is a mirage.

Focus on Bottleneck Activities

- Bottlenecks govern both throughput and inventory.
- The transfer batch may not, and many times should not be, equal to the process batch.
- A process batch should be variable not fixed.
- Schedules should be established by looking at all of the constraints simultaneously. Lead times are the result of a schedule and cannot be predetermined.

Parallel Routing of Activities

Linear Process

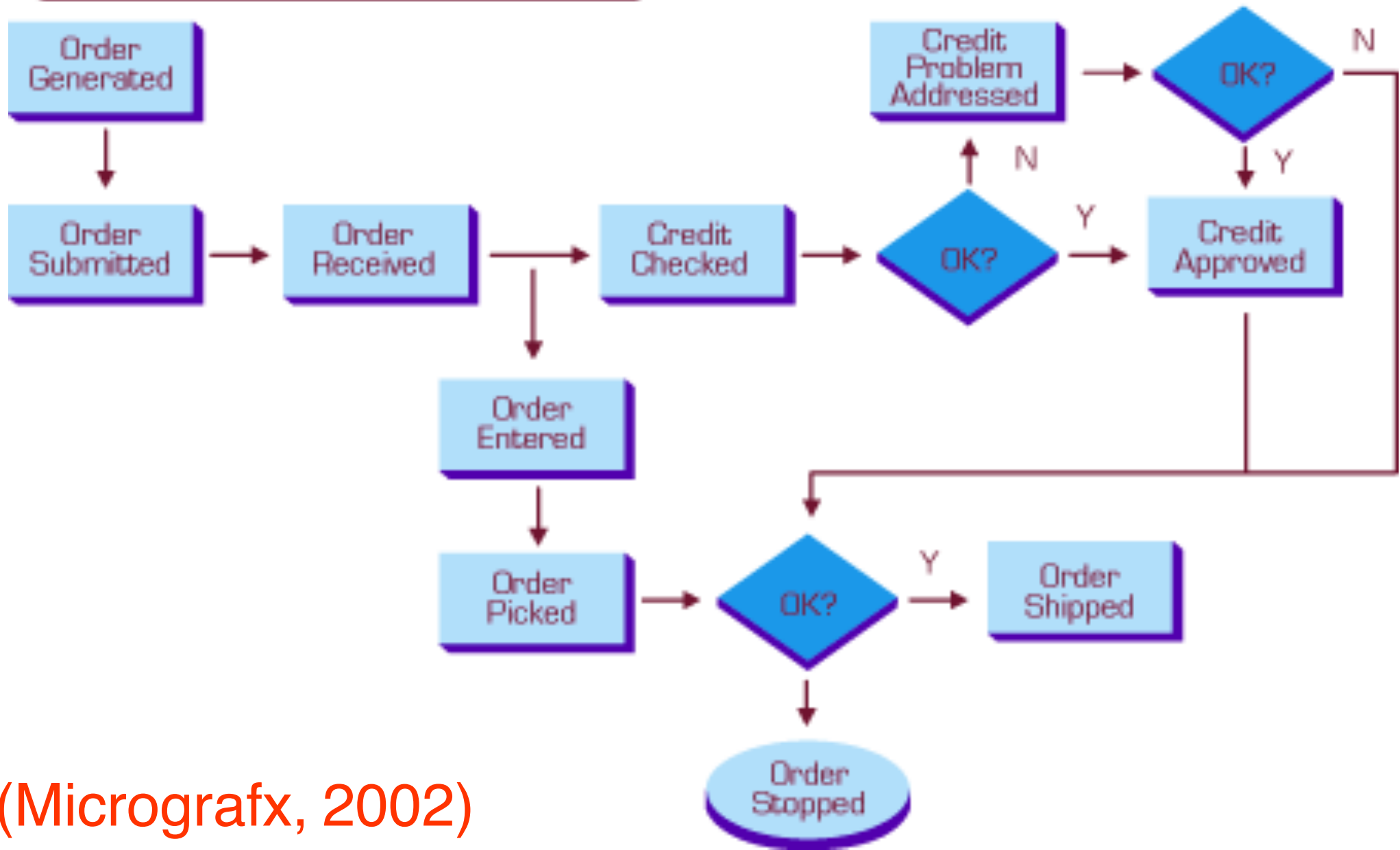


(Micrografx, 2002)

Parallel Routing of Activities

What is the advantage?

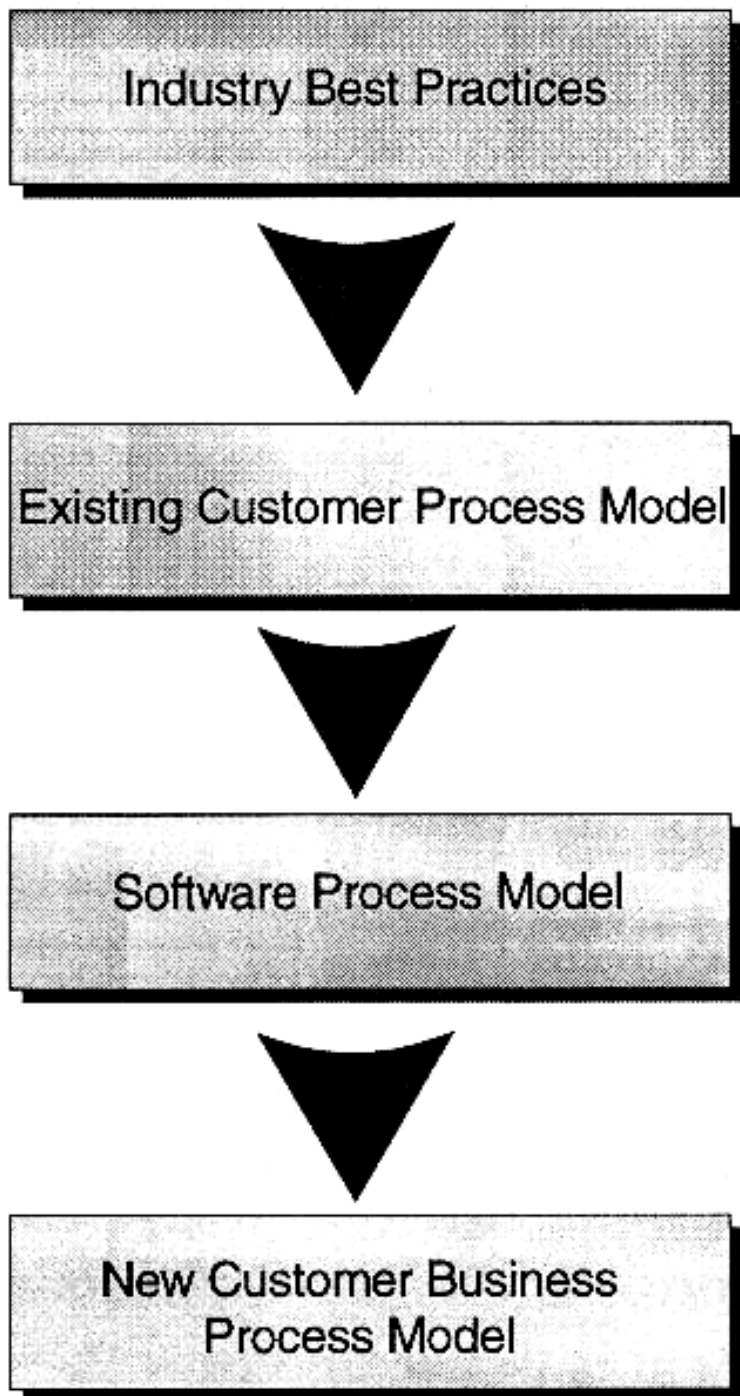
Parallel Process



(Micrografx, 2002)

Resource-related Improvements

- Integration of activities
 - 'Job enlargement' - the integration of two activities on the same level of difficulty
 - 'Job enrichment' - create more challenging activities for the involved employees
- Assignment of Resources



Business Engineering

- Crucial to success of the ES project
- Companies examine their current business process, industry best practices, & the ESs process model & combine them to design a new business process model
- Leverage the business process model as a starting point for maximum automation of systems & configuration customising.

Critical Examination of Processes

Examine

The Purpose	for which
The Place	at which
The Sequence	in which
The Person	by whom
The Means	by which

the activities
are undertaken

with a view to

{	Eliminating	those activities
	Combining	
	Rearranging	{
	Simplifying	

Critical Examination of Processes

❑ Purpose – What & Why

❑ Place – Where & Why

❑ Sequence – When & Why

❑ Person – Who & Why

❑ Means – How & Why

Redesign processes by

- Making sure a company focuses on creating value for customers and suppliers
- Integrating all critical business processes
- Addressing the management of whole processes, not just the management of individual tasks
- Reducing or even eliminating lateral hand-offs or complex chains of business steps

Business Engineering

- "Business engineering is the rethinking of business processes to improve and accelerate the output of processes, materials or services." (Philip Morris, Lausanne, Switzerland)
- "It's the streamlining of business processes to have maximum effect with minimum resources in supporting company goals." (Ernst & Young, South Africa)
- "Business Engineering is the re-thinking of business processes to improve the speed, quality, & output of materials or services." (Philip Morris European Union Region, Switzerland)

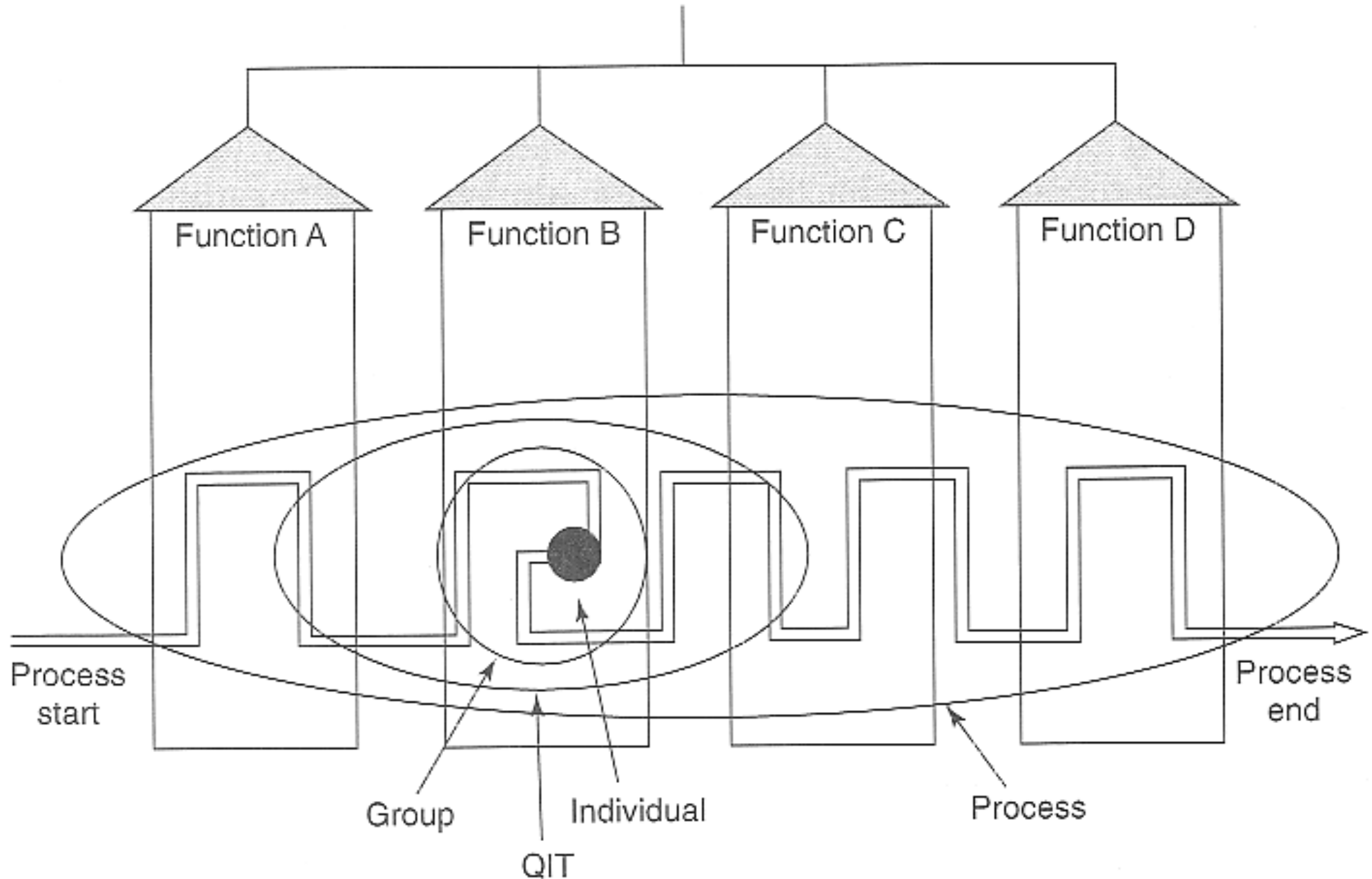
Business Engineering

- "It's the search for an optimal flow in a company" (Messerli AF, Switzerland)
- "Business Engineering revolves around information technology and continuous change. It is the constant refinement of an organization's changing needs." (Esso, Austria)
- "Generally, it's a customer focus. It's also the designing of new processes using new information technology to create an efficient business network that involves creative staff in the process redesign." (Fahrzeugausriistung Berlin GmbH, Germany)

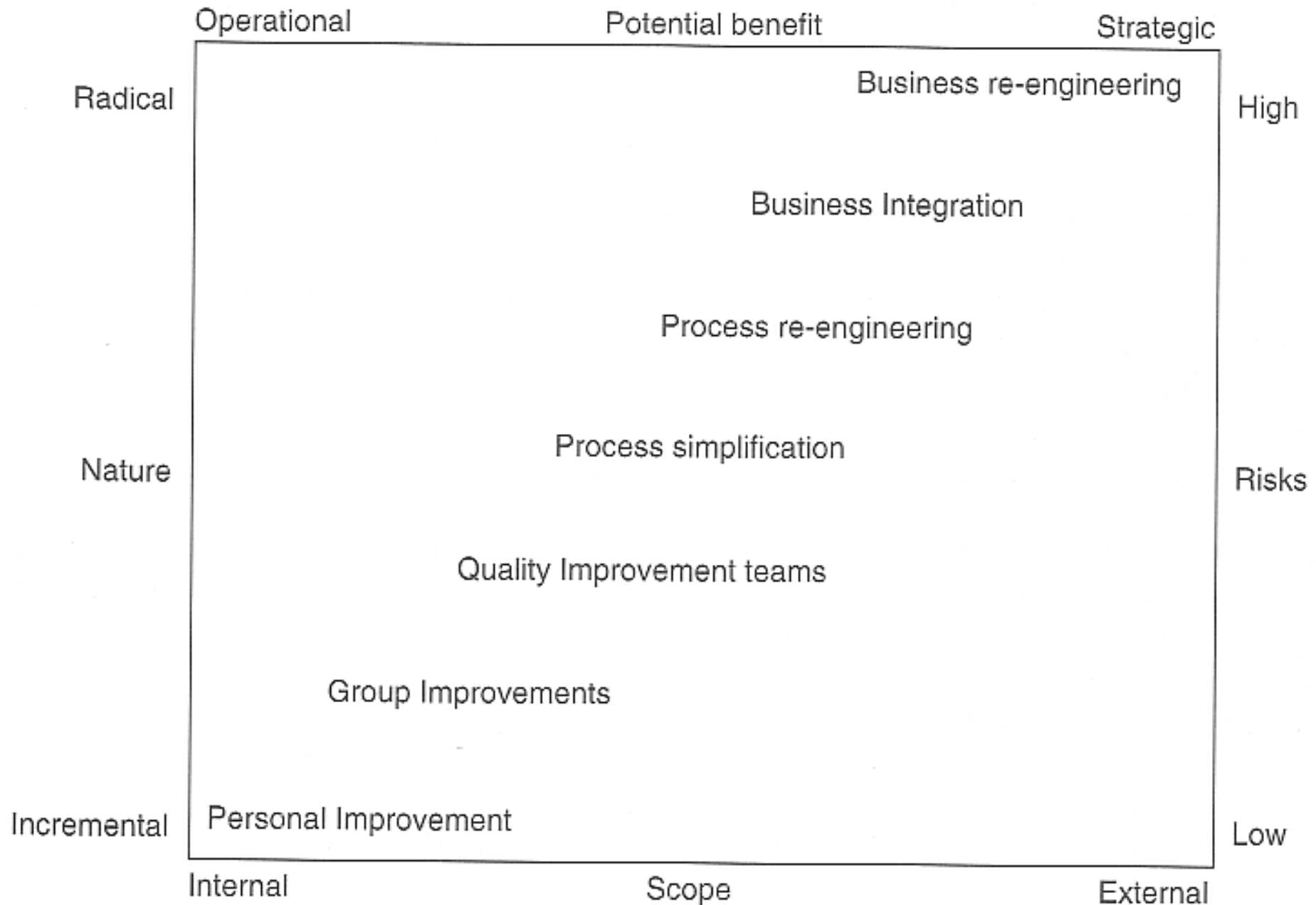
Business Engineering enables Co's

- Rethink and redesign business processes
- Resist the temptation to automate obsolete processes with modern technology
- Stress the importance of simplicity in an increasingly complex world
- Search for new ways to organize work
- Establish change as a constant element of business in today's world
- Recognize and realize the potential of new technology

Processes & Process Initiatives



Comparison of Process Initiatives

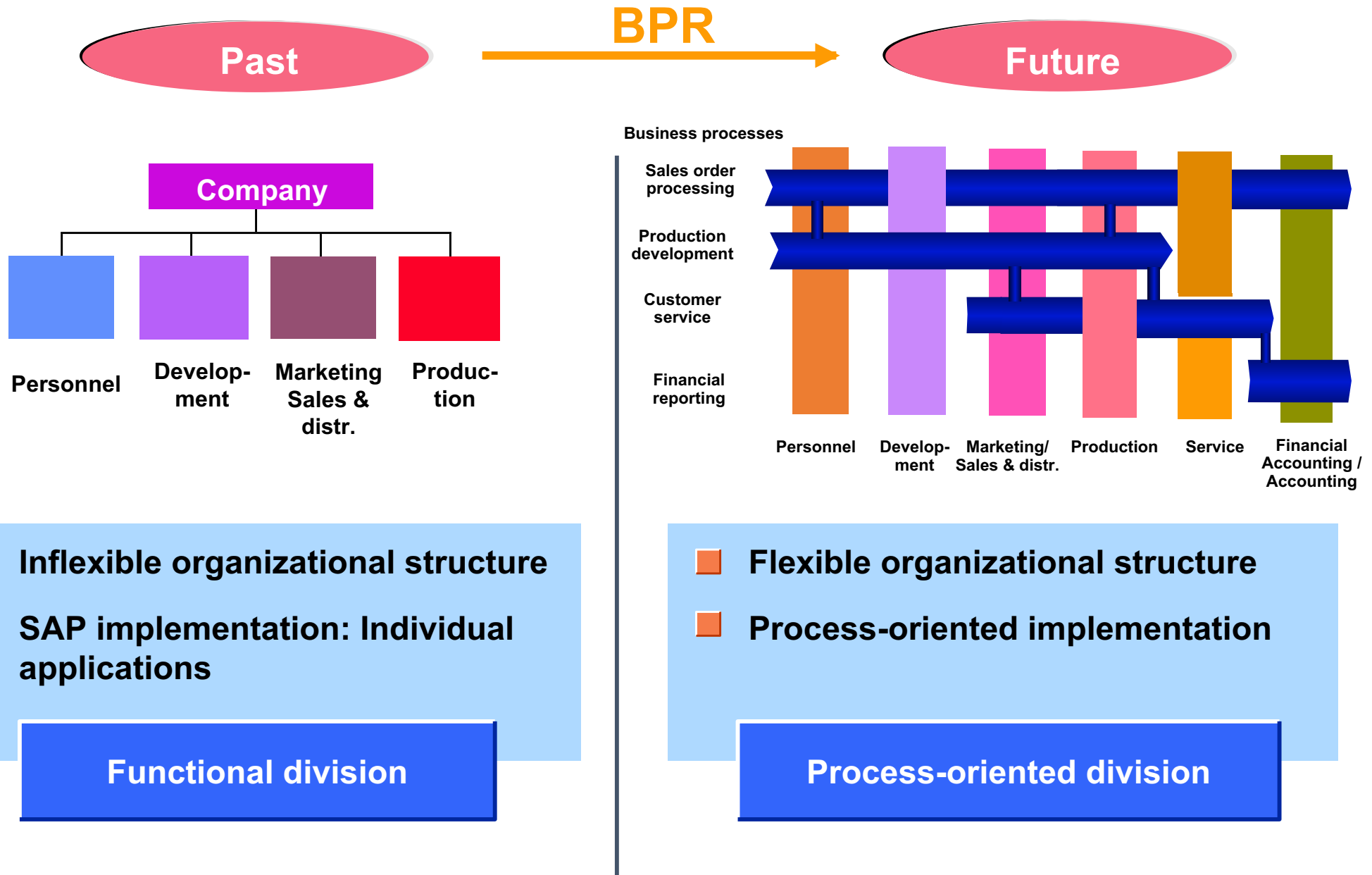


Need, Application, & Mgmt Capability

WHY NOT?	OK	Need BPR
OK	DANGER	Don't Need BPR
Don't Apply BPR	Do Apply BPR	

HOW TO DEVELOP?	OK	Need BPR
OK	DANGER	Don't Need BPR
LOW	HIGH	
MANAGEMENT CAPABILITY FOR BPR		

Evolving Organisational Structures



Finding Opportunity Using BPR

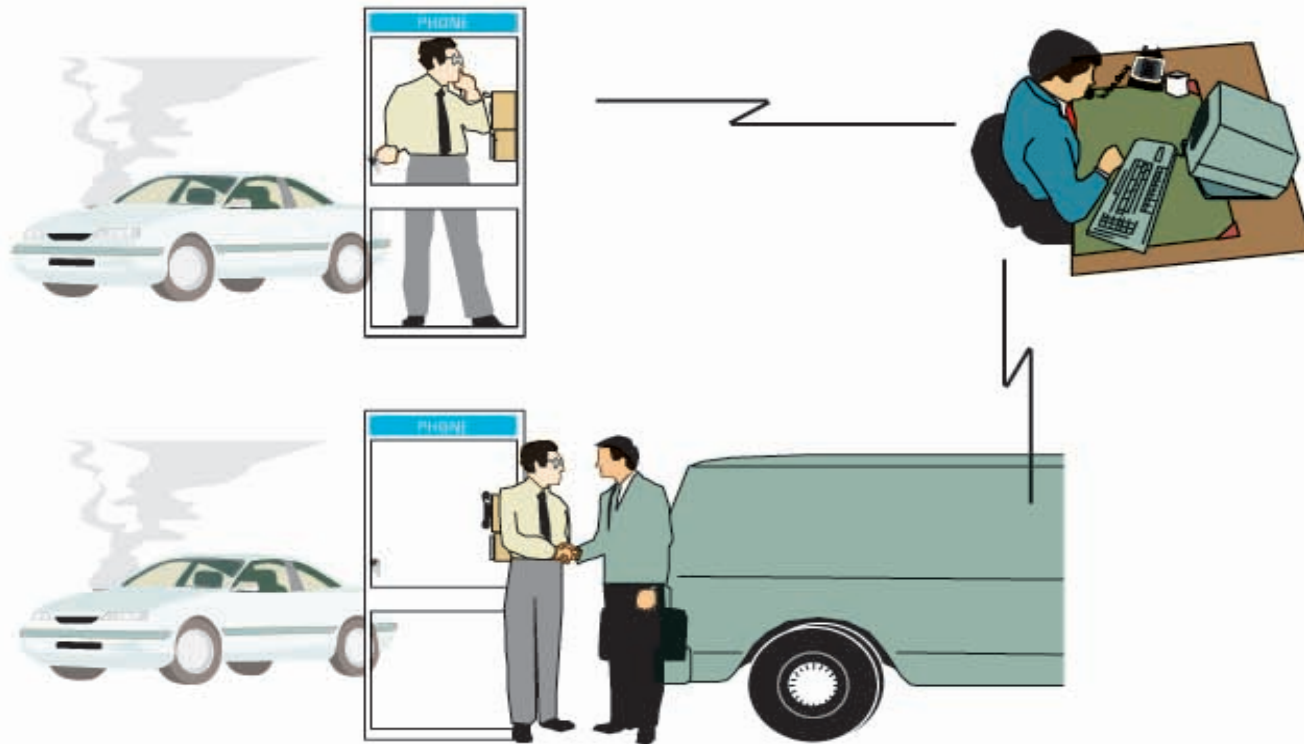
Company A: Claims Resolution Process



Resolution Cycle Time: 3–8 weeks

Finding Opportunity Using BPR

Progressive Insurance: Claims Resolution Process



Resolution Cycle Time 30 min–3 hours

Process Implementation

Process Implementation

- Organizational point of view - the involved staff members must be trained in the new procedures (new objectives, new interfaces, and new policies, hierarchical organizational chart, embedding in cross-functional processes)
- IT point-of-view - process implementation includes the development of new software or the configuration of standard packages like ERP software
- Pragmatise the to-be models

Process Execution

Process Execution

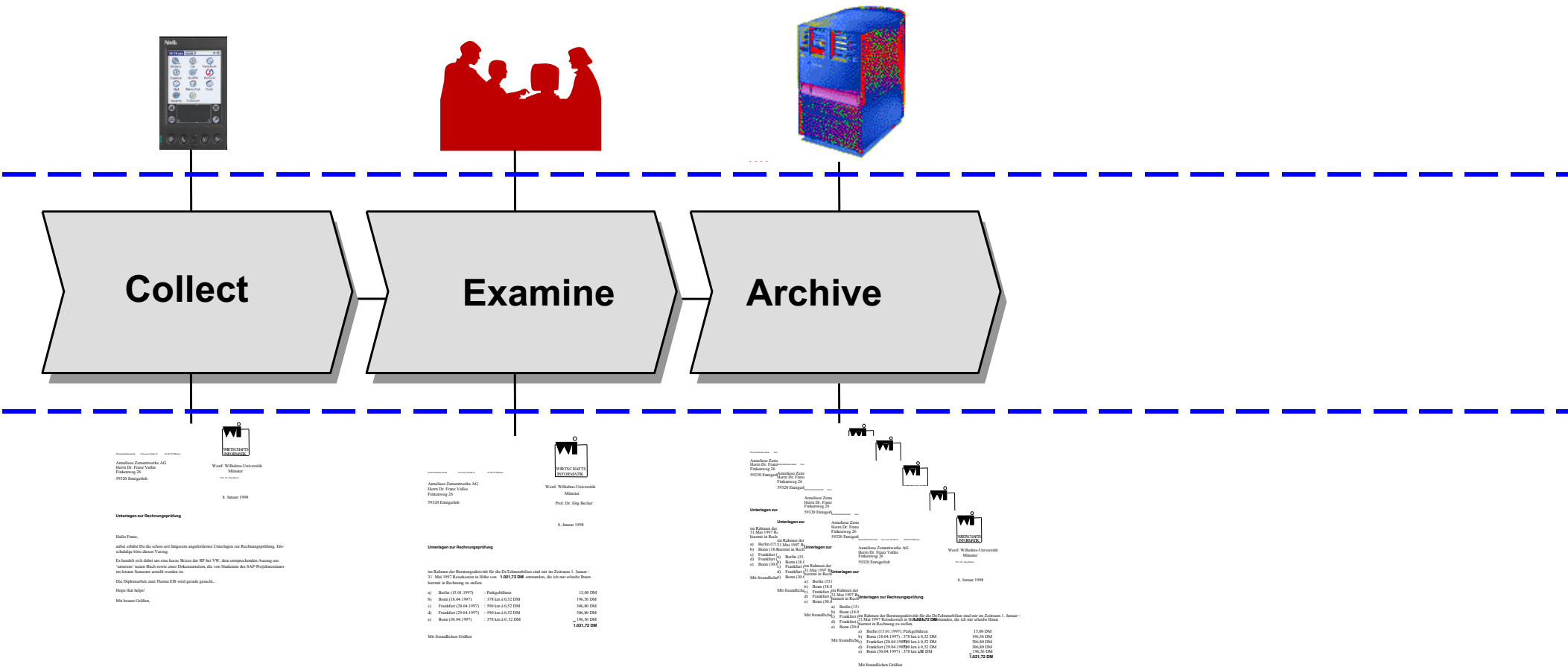
- Realising the benefits
- Document the usage of processes – may lead to modification
- Stabilise and freeze

Process

Monitoring

Controlling

Process Monitoring & Controlling



3 perspectives of Process Controlling

Benchmarking

Rosemann (2001)

Process/Function Perspective

- **Examples of evaluations:**
 - Average processing time
 - Work-time/idle-time-relation
 - Development of the processing time (learning curve)
 - Work-time for a particular order
 - Identification of time- and money-intensive activities (using the activity-based costing approach)

Object Perspective

- Cause-effect relationship between objects and processes
- Identification of *complexity generating activities*
 - harmonization of processes
 - elimination of objects without benefit
- e. g. characteristics of incorrect invoices
 - supplier rankings (ABC-Analysis)
 - statistical checking of invoices
- ABC Costing on objects, time related information on objects, quality indicators

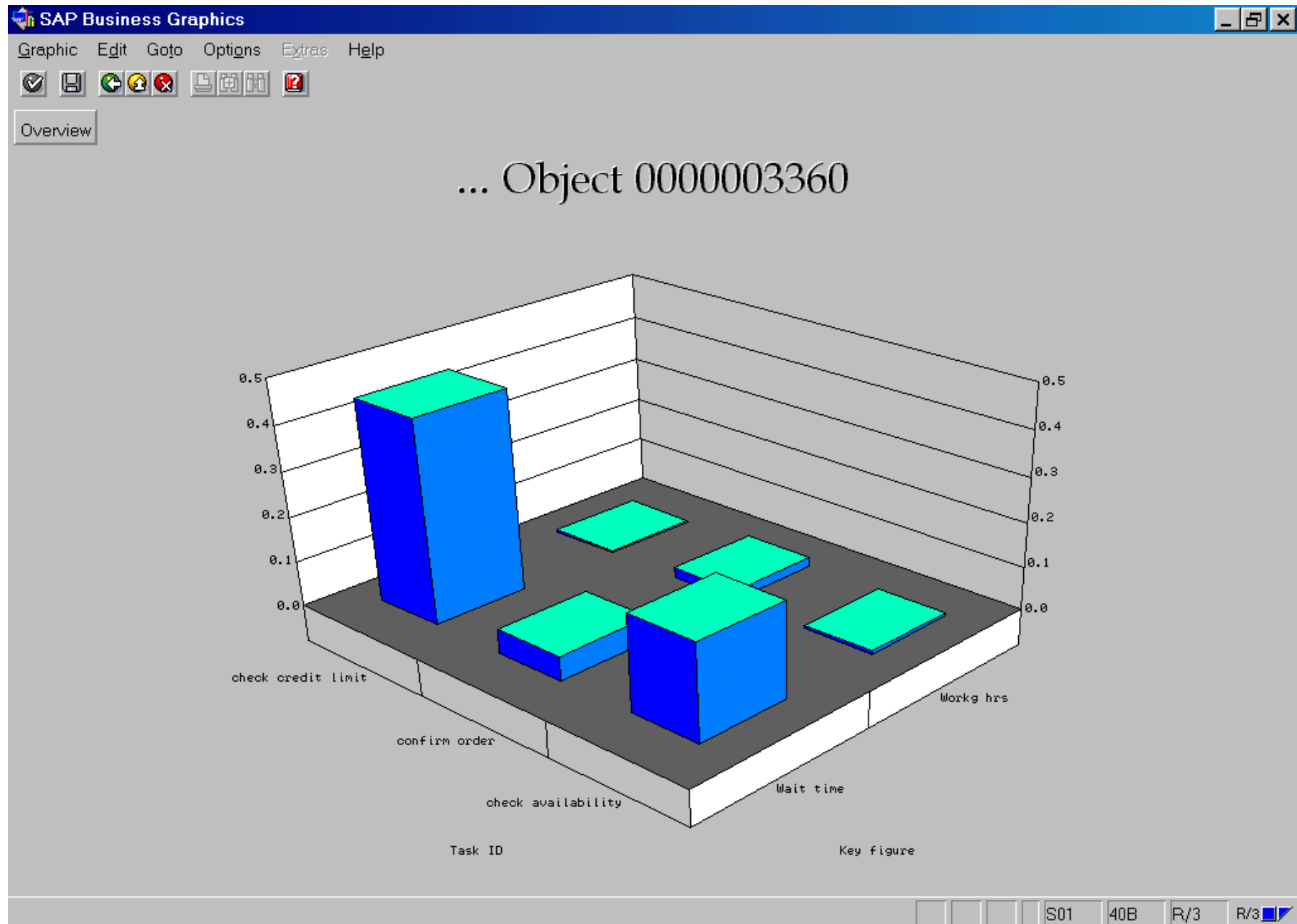
Resource Perspective

- Management of resource interdependencies (e. g. staff resolution based on work loads)
- **Examples of evaluations:**
 - Number of involved (human, technical) resources
 - Resource load factor
 - Qualification adequacy
 - Exception statistics/Triggering of maintenance activities

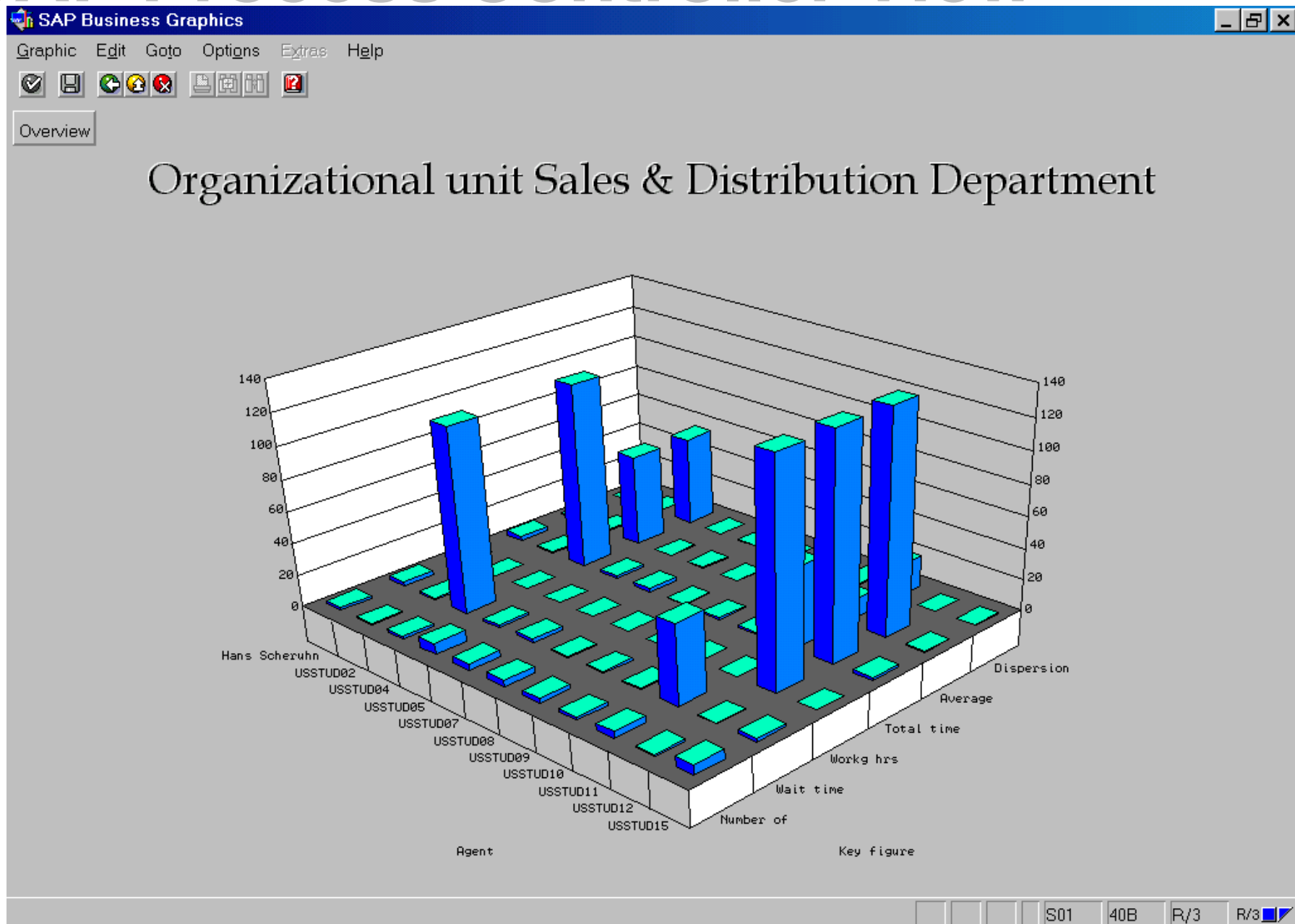
Process Performance measuring (Process Controller view)

- Cycle times of processes and functions
- Frequencies of processes and functions
- Deviations of cycle times and frequencies
 - depending on customers
 - depending on the sales organisation
 - depending on the distribution channel
 - depending on various other dimensions
- Process quality
 - Breaks in organisational flow
 - Number of processing steps
 - Number of involved organisational units
- Process status

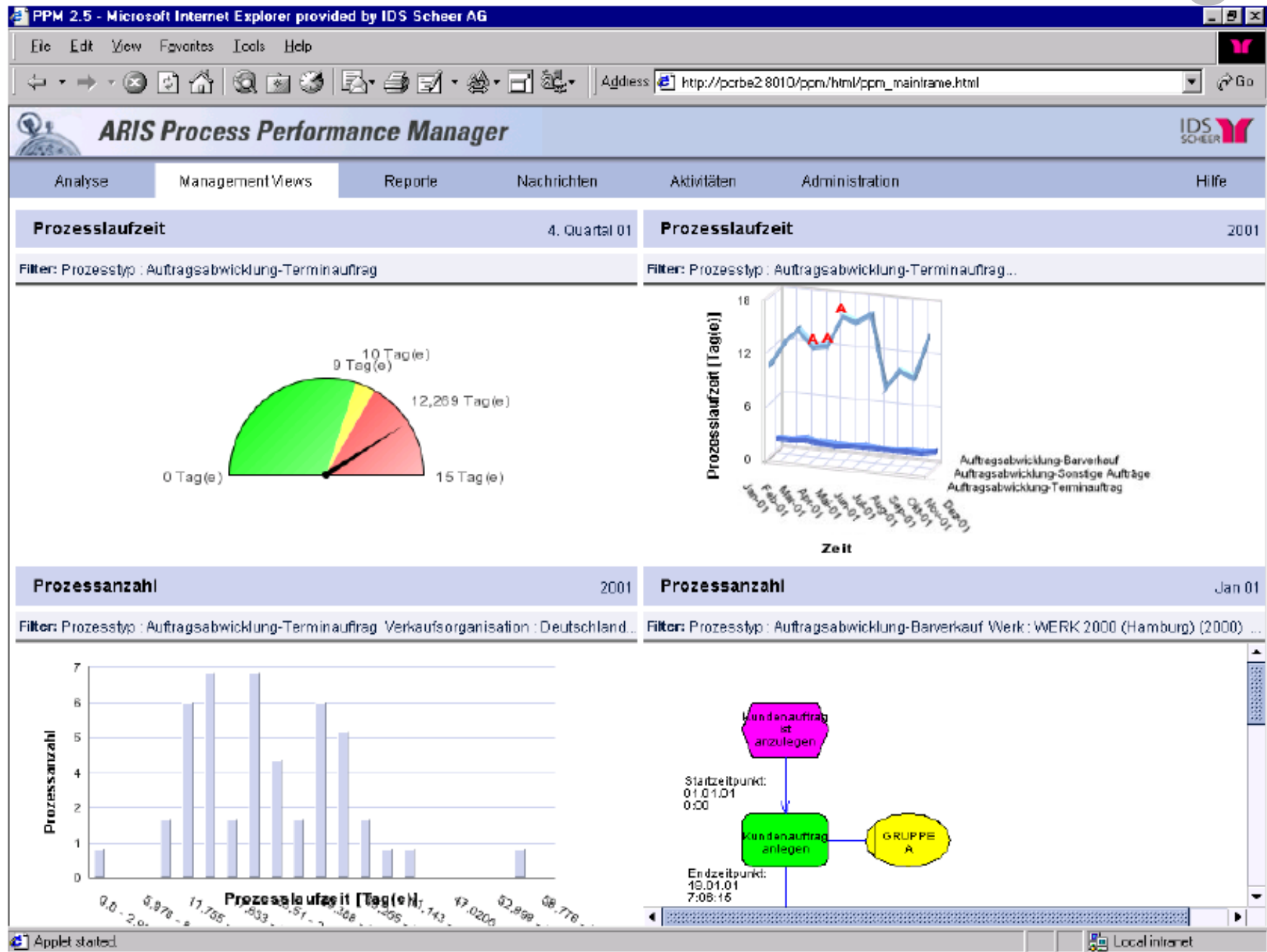
SAP Process Controller View



SAP Process Controller View



ARIS Process Performance Manager



Process Change Management

- ❓ Continuous identification & evaluation of new events (new competitors, changing demands from customers, leaving employees) may lead to changes of the organisation and/or IT systems - Maintain process model and check if the real process still follows the model
- ❓ Implement a process-oriented culture

Why Signavio? Leader for the past 3 years



Gartner's Magic Quadrant for Business Process Mining Tools

